Search Frictions and Welfare in a Market for Prominence: Theory and Experiment*

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Abstract

In online marketplaces, sellers pay platforms to make their products more visible and easier to search, creating a market for search prominence. I study how search friction affects the level and distribution of surplus in this market. In my model, two *ex-ante* identical sellers bid for a prominent position in a second-price auction and then simultaneously post prices. The consumer observes both prices and learns her match value for the top seller's product for free before deciding whether to incur a search cost to search the second seller's product. I derive equilibrium bids, price distributions, and search cutoffs, and test these predictions in a laboratory experiment with low and high search costs. Platform auction revenue changes little as search cost rises, despite the predicted increase. Total and consumer surplus decrease with search cost, but consumer surplus falls proportionally much less than predicted. At both search costs, consumer surplus exceeds equilibrium benchmarks, whereas sellers jointly receive only about a fifth of total surplus after paying for prominence. Further analysis of bidding, pricing, and search behavior helps explain these findings.

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1 Introduction

Online marketplaces offer a wide range of products, but these products are not displayed equally prominently. Some appear at the top of search results or receive more visible display space, while others require consumers to scroll through several pages of results. Sellers pay platforms to make their products easier to discover and evaluate, creating a market for search prominence. Google search ads, Amazon Sponsored Products, and sponsored hotel listings are familiar examples. These arrangements give sellers a way to compete for consumers' attention in addition to competing for their purchases. The return to that payment depends on whether consumers search the seller's product and what they do after learning about it.

Consumers spend time and effort learning whether a product suits their needs and may stop before searching every alternative. A top seller can therefore reach buyers who never search its competitors' products. The value of this advantage depends on how costly those competing products are to evaluate. If learning about an alternative requires substantial effort, a consumer may purchase the product already considered even though another product could offer a better match. Prominence consequently has value even when it conveys no information about product quality. It changes the cost of reaching a seller relative to its competitors and thereby changes the demand that seller can expect to receive.

Readily available price information also allows sellers to influence consumers' search decisions. Prices appear in product listings and price advertisements, so consumers can often compare them before undertaking a detailed search. Learning whether a product is a good match can nevertheless remain costly. By posting a price, a seller changes both the surplus from purchasing its product and the consumer's incentive to search another product. A low price at a competing seller may make further search worthwhile, whereas an attractive price at an already considered seller may encourage the consumer to stop. Prices thus help determine the products that enter consumers' consideration sets, as well as their choices within those sets. The value of prominence depends on how sellers use these publicly observed prices to compete when search is costly.

This paper asks how search friction affects the level and distribution of surplus in a market where sellers compete for prominence and prices are visible before search. I develop a model linking bids for a prominent position to subsequent pricing and consumer search, and test its predictions in a laboratory experiment. The experiment brings these decisions together in one market, allowing me to observe sellers' bids for prominence alongside the prices and buyer choices that determine its return. It also measures platform revenue, consumer surplus, and seller profits within the same market. This distinguishes changes in the surplus created through trade and search from transfers among consumers, sellers, and the platform. A valuable position need not leave its winner with a large profit after payment, and an increase in search cost can affect both the value of the position and the surplus available to divide.

In field settings, prominence can combine a search-cost advantage with information about quality or relevance, making these channels difficult to disentangle. Personalized recommendations can also convey information about a consumer's likely match. The laboratory allows me to separate the search-cost advantage from these informational channels. A prominent product may attract attention both because it is easier to find and because consumers expect it to be more suitable. In the experiment, sellers have identical distributions of match values and do not know their realized matches when they bid or price. Prominence allows the buyer to learn the top seller's match value for free, while learning the second seller's match value is costly. Varying that cost while holding the remaining market rules fixed changes the search advantage without changing what the position reveals about product fit.

The model has two ex-ante identical sellers and a buyer with unit demand. Sellers first compete in a second-price auction for the top position and then simultaneously post prices. Each product is independently a good or bad match for the buyer. The buyer observes both prices and learns her match value for the top seller's product for free. She decides whether to pay to search the second seller's product and can buy a product whose match value she has learned or exit without purchasing. Sellers anticipate the resulting demand when they price, and anticipate the revenue difference between the two positions when bidding for prominence. The value of the auctioned position therefore emerges from competition in the product market.

The buyer's incentive to search depends on her initial match. After a good match, further search offers a chance to obtain a suitable product at a lower price. The difference between the two prices determines whether the expected saving justifies the search cost. After a bad match, the buyer instead searches for a product worth buying, so the relevant price is the second seller's. The top seller can use its price to retain a buyer who has found a good match. The second seller's price affects the incentive to search its product after either initial match. Public prices consequently link competition for purchases to competition over whether search continues.

I derive an equilibrium that connects these search incentives to pricing, bidding, and welfare. The top seller charges more on average than the second seller, reflecting its advantage in reaching the buyer without requiring costly search. In equilibrium, the expected revenue advantage of the top position equals the search cost. The auction then has a symmetric equilibrium in which both sellers bid that amount, so platform revenue increases one for one with search cost. Payment for prominence eliminates the top seller's expected revenue advantage in profits, leaving the two sellers with equal expected profits. These predictions connect the platform's ability to collect revenue to the pricing and search decisions that make prominence valuable.

The experiment implements the game at a low and a high search cost. Subjects make bidding, pricing, search, and purchase decisions in repeated markets, with roles fixed and

participants randomly rematched between rounds. The treatment changes the cost of learning the second seller's match value rather than assigning a different value directly to the auctioned position. Sellers must assess how this change affects the subsequent product market when deciding how much to bid. Observing both stages makes it possible to examine whether the predicted increase in position value appears in auction payments and whether buyers and sellers receive the predicted shares of surplus.

The welfare findings show substantial departures from these linked predictions. Platform revenue changes little as search cost rises, despite the model's predicted increase. Average bids rise by about 4.8 points rather than the predicted 20 points, and the response of payments is smaller still. In the second-price auction, the losing bid determines the top seller's payment to the platform. An increase in the winning bid need not therefore translate into an increase in platform revenue. The observed weak response of payments accompanies a much smaller response of bids than the equilibrium change in the value of prominence.

Consumers receive more surplus than predicted at both search costs. Consumer and total surplus are lower in the high-cost treatment, but consumer surplus falls proportionally much less than the equilibrium benchmark suggests. This does not mean that the market creates more surplus than predicted. Total surplus falls below its benchmark in both treatments, while lower payments to sellers allow consumers to retain more of the surplus that is created. The experiment thus separates two departures from equilibrium that would be obscured by examining consumer surplus alone. The allocation creates less total surplus, yet its distribution is more favorable to consumers.

Sellers jointly retain about one fifth of total surplus after paying for prominence at either search cost. Their revenues fall below the equilibrium benchmark in both treatments, but the implications for profits differ. At the low cost, payments above the predicted level add to the revenue shortfall, leaving combined profits well below equilibrium. At the high cost, payments far below the predicted level largely offset the revenue shortfall, so combined profits are close to the benchmark. Similar observed profit shares therefore coexist with different departures in revenues and payments. The auction transfer is central to understanding how product-market performance translates into seller profits.

Pricing and search behavior help account for these distributional findings. Top sellers charge more on average than second sellers, but their prices are substantially below equilibrium predictions at both costs. Prices at both positions are also more dispersed than predicted, with observations beyond the equilibrium supports. At the high search cost, second sellers charge more than predicted. Buyers with a good initial match largely follow the search cutoff, so lower top prices can benefit them through both lower purchase payments and a reduced incentive to incur further search costs. After a bad initial match, high second prices can make stopping optimal even when the unsearched product would turn out to be a good match. Such outcomes reduce the gains from trade without requiring a mistake by the buyer. Buyers

also depart from the search rule at the observed prices, with both too much and too little search relative to private optimality.

The distinction between revenues and profits also matters for the return to prominence. In the data, the top position delivers higher average revenue at both search costs. Its average profit is nevertheless below that of the second position at the low cost and above it at the high cost. Comparing these outcomes with bids and payments shows how competition for the position absorbs part of its product-market return. These comparisons describe the outcomes of the auction-assigned positions. They do not identify the causal gain from moving a particular seller to the top or equate a realized revenue difference with the expected return relevant when a seller submits its bid. Together with the pricing and search evidence, they connect the welfare findings to the market decisions through which search friction affects the value of prominence.

1.1 Related literature

The paper builds on models in which search frictions affect price competition. Diamond (1971) shows how costly price information can sustain monopoly pricing. With differentiated products, search also allows consumers to find a better match, as in Wolinsky (1986) and Anderson and Renault (1999). When prices are observable before search, they can influence which products consumers search and whether search continues. Choi et al. (2018), Ding and Zhang (2018), and Obradovits and Plaickner (2023) study pricing under this information structure. The present model retains costly learning about product fit with publicly observed prices, but sellers acquire the top position through an auction. A low price by the second seller can induce continued search, but does not move that seller ahead of the initial product information supplied by prominence. This separates the acquisition of the position from the subsequent use of prices to influence search, while allowing both decisions to respond to the same friction.

Prominence and ordered-search models explain how an advantage in reaching consumers changes price competition. Armstrong et al. (2009), Zhou (2011), and Armstrong (2017) study pricing when consumers encounter firms in a particular order. Armstrong and Zhou (2011) examine how firms obtain prominence, including through payments to intermediaries and competition in advertised prices. These papers connect the value of being considered early to subsequent pricing and consumers' continuation decisions. Field evidence also supports distinguishing search from purchase. Using hotel-ranking variation, Ursu (2018) provides evidence for a search-cost channel through which rankings affect consumer search. My focus is how the cost of searching competing products affects the value of a purchased search advantage and the distribution of surplus. The experiment makes the difference in search costs explicit and holds sellers' match distributions fixed, so the winning position does not signal better product quality or a better match.

Competition for prominence also connects the paper to sponsored-search auctions. Edelman et al. (2007) and Varian (2007) analyze auctions allocating advertising positions with different click volumes. Chen and He (2011) and Athey and Ellison (2011) incorporate consumer search and the information conveyed by paid rankings. The closest theoretical comparisons for endogenous position values are Xu et al. (2011) and Anderson and Renault (2026). Xu et al. (2011) derive position values from subsequent price competition, while Anderson and Renault (2026) show how incentives to move in the search order depend on competition from firms ahead of and behind them. I bring this connection into the laboratory by observing bids together with the pricing and search decisions that generate the position's return. The second-price auction implements competition for the search advantage, allowing the experiment to examine how its value responds to search cost.

Experimental studies of search and pricing provide the basis for examining the behavior that generates these position values. Davis and Holt (1996), Abrams et al. (2000), and Cason and Friedman (2003) test how search and information frictions affect prices and market performance. Brown et al. (2018) examine sellers' use of exploding offers to discourage further search when prices are visible. Their experiment shows why consumers' responses matter for the profitability of a strategy intended to prevent continued search. The present experiment introduces competition for prominence into a market with endogenous pricing and consumer search. Sellers learn their auction-assigned positions before pricing, buyers can return to a previously searched product, and the treatment varies the cost of searching the second product. Observing the auction and product market together permits a welfare comparison that includes the payment for prominence, the prices consumers pay, and the search costs they incur.

The remainder of the paper is organized as follows. Section 2 presents the theoretical framework. Section 3 describes the experimental design and hypotheses. Section 4 presents the welfare findings and examines bidding, pricing, search, and returns to prominence. Section 5 examines individual decisions and adjustment over time. Section 7 concludes. The appendices contain proofs, further analyses, and the experimental instructions.

2 Theoretical Framework

2.1 Model

There are two ex-ante identical sellers and one buyer with unit demand. Each seller offers one product at zero marginal cost. The buyer's match values are independent across sellers, with

$$v_k = \begin{cases} V & \text{with probability } q, \\ 0 & \text{with probability } 1-q, \end{cases} \qquad k = 1, 2, \tag{1}$$

where $V > 0$ and $0 < q < 1$. The cost of searching the second seller's product satisfies $0 < s < qV$. The timing, information structure, payoff functions, match distribution, and search cost are common knowledge. Realized match values are not public information. Sellers do not observe them when bidding or pricing, and the buyer learns each product's match value only when she searches that product.

The game has three stages, summarized in Figure 1. Stage 1 is the market for prominence. Stages 2 and 3 form the product market.

Stage 1. Auction for prominence. Sellers simultaneously submit bids in a second-price auction. The higher bidder receives the top position and pays the losing bid, $b^{(2)}$. The other seller receives the second position and pays nothing. If bids are equal, each seller has probability one-half of receiving the top position. After the auction, subscripts identify positions rather than seller identities. Seller 1 is the top seller and seller 2 is the second seller.

Stage 2. Pricing. After learning their positions and the position payment, sellers simultaneously post prices $p_1, p_2 \geq 0$.

Stage 3. Search and purchase. The buyer observes both prices and searches the top seller's product for free, learning v_1. She may pay s to search the second seller's product and learn v_2. Before searching the second seller, she may buy from the top seller or exit. After searching the second seller, she may buy from either seller or exit. Returning to the top seller is free, and buying from the second seller requires search.

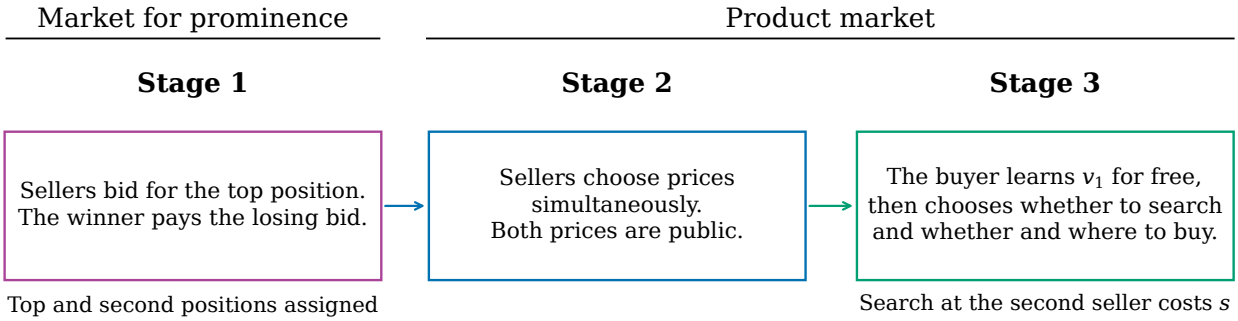


Figure 1: The three stages of the game

The auction assigns positions in the market for prominence. Sellers then post prices and the buyer makes search and purchase decisions in the product market.

Let $x \in \{0, 1\}$ indicate whether the buyer searches the second seller and let $a \in \{0, 1, 2\}$ denote her purchase decision, with $a = 0$ denoting exit. The buyer's payoff is

$$u(a, x) = \sum_{k=1}^{2} (v_k - p_k) \cdot \mathbf{1}_{\{a=k\}} - s \cdot x. \tag{2}$$

Seller revenues, seller profits, and platform revenue are

$$\Pi_k = p_k \cdot \mathbf{1}_{\{a=k\}}, \qquad \pi_1 = \Pi_1 - b^{(2)}, \qquad \pi_2 = \Pi_2, \qquad \text{PR} = b^{(2)}. \tag{3}$$

The position payment is sunk when sellers choose prices.

I solve the game backward. The buyer's search and purchase decisions determine each seller's demand at a given price pair. These demand functions yield a pricing equilibrium for the product market. The difference between the resulting expected revenues at the two positions determines the value of prominence and the sellers' auction bids.

2.2 Buyer search and seller demand

After observing (v_1, p_1, p_2), the buyer compares the expected gain from searching the second seller with its cost. With a bad top match, search offers a chance of finding a product worth buying. With a good top match, it offers a chance of buying a suitable product at a lower price. The following lemma specifies optimal choices, including choices at indifference.[1]

Lemma 1 (Optimal search cutoffs). *For $0 \le p_1, p_2 \le V$, the buyer's optimal search decision satisfies*

$$x^*(0, p_1, p_2) = 1 \quad \Longleftrightarrow \quad p_2 \le V - \frac{s}{q}, \tag{4}$$

$$x^*(V, p_1, p_2) = 1 \quad \Longleftrightarrow \quad p_1 - p_2 > \frac{s}{q}. \tag{5}$$

Thus, at a search tie the buyer searches after a bad top match and stops after a good top match. She buys an available product with the largest nonnegative purchase surplus or exits if all available products have negative purchase surplus. At ties, purchase is preferred to exit and the top seller is preferred to the second seller.

Proof. See Appendix A.2, which also specifies optimal behavior after prices outside $[0, V]$.□

Figure 2 summarizes the buyer's choices. The complete buyer decision tree is in Appendix A.1.

For $0 < p_1 \le V$ and $0 < p_2 \le V - \frac{s}{q}$, let $D_k(p_1, p_2)$ be seller k's sale probability. The search cutoffs imply

	$D_1(p_1, p_2)$	$D_2(p_1, p_2)$
$p_1 - p_2 > \frac{s}{q}$	$q(1-q)$	q
$p_1 - p_2 \le \frac{s}{q}$	q	$q(1-q)$.

(6)

When a buyer with a good top match stops, the top seller serves all such buyers, while the second seller sells only when the top match is bad and the second match is good. When the

[1] The tie rules support the pricing equilibrium derived below. Appendix A.1 states them formally.

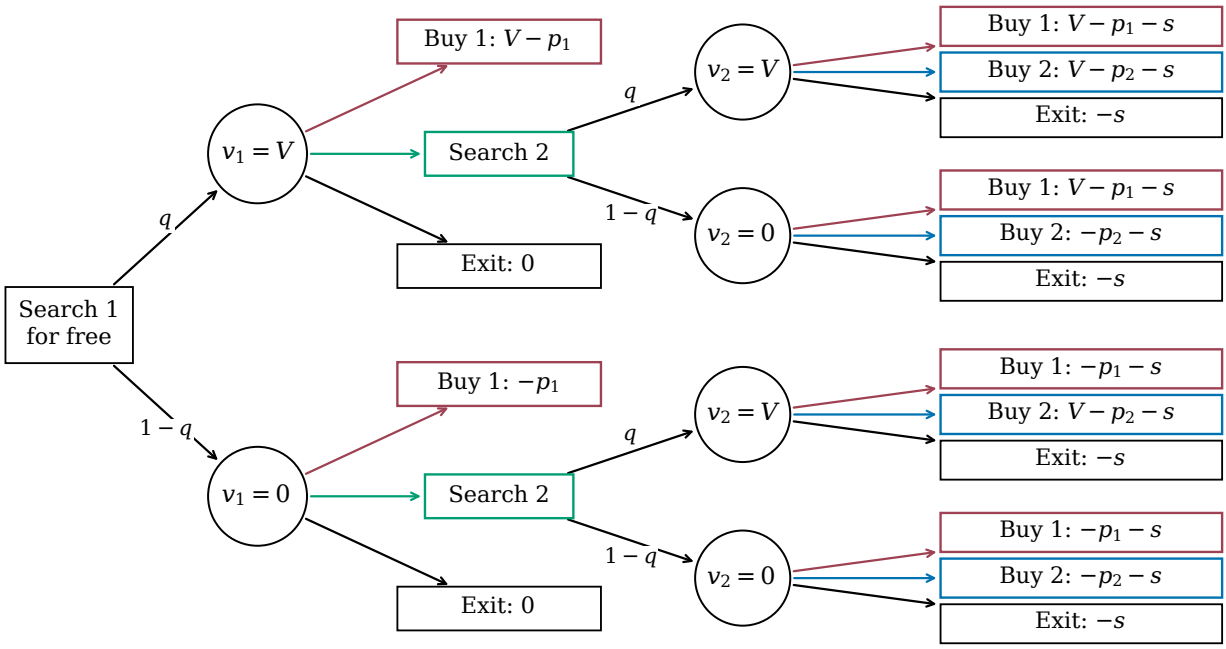


Figure 2: The buyer's search and purchase decisions

The buyer observes both prices and learns the top match for free. Search of the second seller costs s. Circles show realized match values and rectangles show choices.

price gap induces search, the second seller also attracts buyers with two good matches. The top seller then sells only when the top match is good and the second match is bad. Figures 3 and 4 show the purchase paths in the two cases. At a zero price, the purchase tie rule can change unit demand, but the zero-price seller's revenue is always zero.

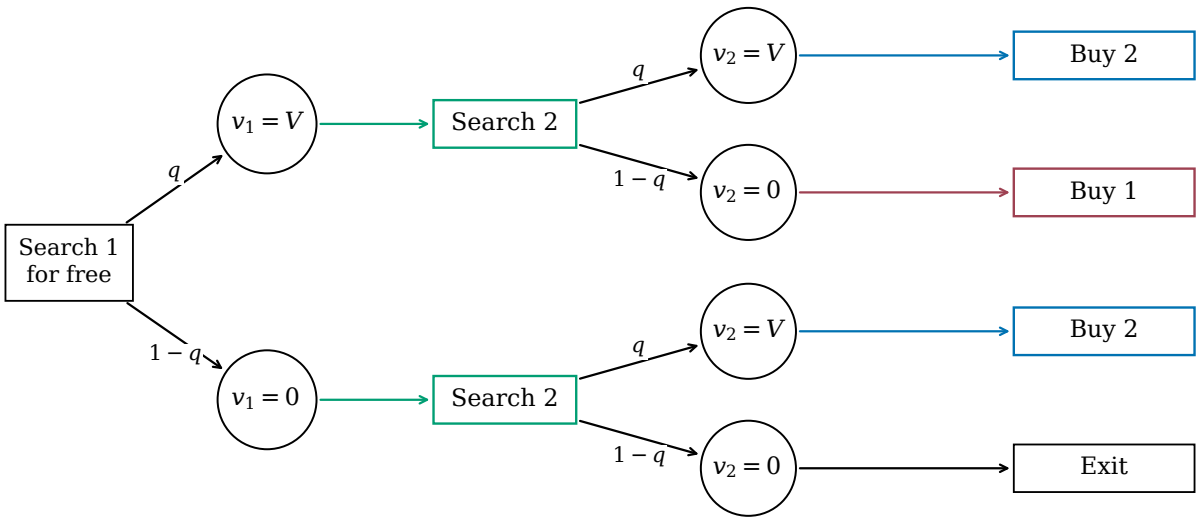


Figure 3: Search after a good top match when $p_1 - p_2 > \frac{s}{q}$

For $0 < p_1 \leq V$ and $0 < p_2 \leq V - \frac{s}{q}$, the buyer searches the second seller after either top match.

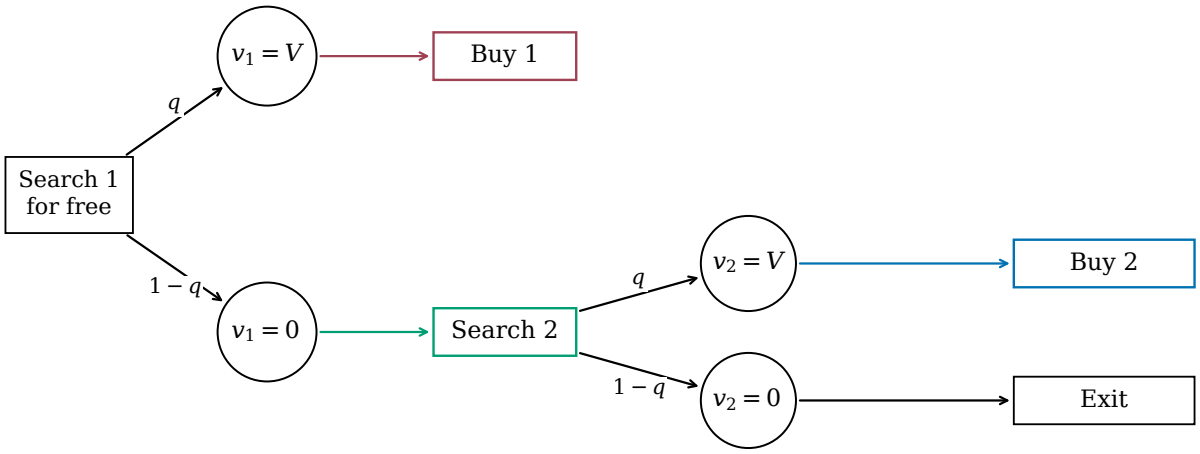


Figure 4: No search after a good top match when $p_1 - p_2 \leq \frac{s}{q}$

For $0 < p_1 \leq V$ and $0 < p_2 \leq V - \frac{s}{q}$, the buyer stops after a good top match and searches the second seller after a bad top match.

2.3 Pricing equilibrium

Proposition 1 (No pure-strategy Nash equilibrium). *Given the buyer behavior in Lemma 1 and $0 < s < qV$, the pricing subgame has no pure-strategy Nash equilibrium.*

Proof. See Appendix B.1. □

On a constant-demand segment, each seller benefits from raising its price. At the search cutoff, the second seller can instead gain demand by slightly lowering its price. These opposing incentives motivate mixed pricing strategies. At deviation prices outside the domain in (6), demand follows the buyer's optimal strategy.

Definition 1 (Mixed-strategy pricing equilibrium). We seek distribution functions $F_1(p_1)$ and $F_2(p_2)$ with connected supports $[\underline{p}_1, \overline{p}_1]$ and $[\underline{p}_2, \overline{p}_2]$ such that

$$p_1^* \in \arg\max_{p_1 \in [0,\infty)} p_1 \mathbb{E}_{F_2}[D_1(p_1, p_2)], \qquad \forall p_1^* \in [\underline{p}_1, \overline{p}_1], \tag{7}$$

$$p_2^* \in \arg\max_{p_2 \in [0,\infty)} p_2 \mathbb{E}_{F_1}[D_2(p_1, p_2)], \qquad \forall p_2^* \in [\underline{p}_2, \overline{p}_2]. \tag{8}$$

Theorem 1 (A mixed-strategy pricing equilibrium). *Given the buyer behavior in Lemma 1 and $0 < s < qV$, the following independent distributions constitute an equilibrium of the pricing subgame:*

$$F_1(p_1) = \frac{1}{q}\left[1 - \frac{(1-q)\left(V - \frac{s}{q}\right)}{p_1 - \frac{s}{q}}\right], \qquad \underline{p}_1 \le p_1 \le \overline{p}_1, \tag{9}$$

$$F_2(p_2) = \frac{1}{q}\left[1 - \frac{(1-q)V + s}{p_2 + \frac{s}{q}}\right], \qquad \underline{p}_2 \le p_2 < \overline{p}_2, \tag{10}$$

where

$$\begin{aligned} \underline{p}_1 &= (1-q)V + s, & \overline{p}_1 &= V, \\ \underline{p}_2 &= (1-q)\left(V - \frac{s}{q}\right), & \overline{p}_2 &= V - \frac{s}{q}. \end{aligned} \tag{11}$$

Both distributions are zero below their lower bounds and one at or above their upper bounds. The top seller's distribution is atomless. The second seller places mass $\frac{s}{qV}$ at $\overline{p}_2$.

Proof. Appendix B.2 verifies both optimality conditions against all nonnegative price deviations. □

The support bounds satisfy

$$\underline{p}_1 - \underline{p}_2 = \overline{p}_1 - \overline{p}_2 = \frac{s}{q}. \tag{12}$$

A higher search cost raises the lower bound of the top seller's price distribution and lowers both bounds of the second seller's distribution. To induce search, the second seller must offer a larger price reduction relative to the top seller.

2.4 Bidding for prominence

Theorem 2 (Symmetric bidding equilibrium). *Suppose the sellers use the pricing equilibrium in Theorem 1 after every auction outcome. Their expected revenues before any position payment are*

$$\mathbb{E}[\Pi_1] = q[(1-q)V + s], \qquad \mathbb{E}[\Pi_2] = q(1-q)\left(V - \frac{s}{q}\right). \tag{13}$$

The value of prominence is their difference,

$$\mathbb{E}[\Pi_1] - \mathbb{E}[\Pi_2] = s. \tag{14}$$

The symmetric bids $b_1 = b_2 = b^* = s$, together with these pricing strategies and optimal buyer behavior, constitute a subgame-perfect equilibrium. Platform revenue and expected seller profits are

$$\mathrm{PR} = s, \qquad \mathbb{E}[\pi_1] = \mathbb{E}[\pi_2] = (1-q)(qV - s). \tag{15}$$

Proof. See Appendix C.1. □

The auction payment absorbs the top seller's expected revenue advantage. A higher search cost raises the value of prominence and the equilibrium payment to the platform, while reducing both sellers' expected profits.

2.5 Welfare

The purchase decision is $a = 0$ for exit and $a = 1$ or $a = 2$ for purchase from the corresponding seller. The search indicator is $x = 1$ if the buyer searches the second seller and $x = 0$ otherwise. Consumer surplus is the buyer's payoff. Total surplus includes both sellers and the platform:

$$\mathrm{CS} = \sum_{k=1}^{2} (v_k - p_k) \cdot \mathbf{1}_{\{a=k\}} - s \cdot x, \tag{16}$$

$$\mathrm{TS} = \mathrm{CS} + \pi_1 + \pi_2 + \mathrm{PR} = \sum_{k=1}^{2} v_k \cdot \mathbf{1}_{\{a=k\}} - s \cdot x. \tag{17}$$

Prices and position payments are transfers. Search uses real resources, so total surplus depends on the product purchased and the search cost incurred. Search can increase surplus

by revealing a valuable second product, but it can also incur a cost without changing the product ultimately purchased.

3 Experimental Design

3.1 Protocol

The experiment implements the three-stage market described above. Each market consists of two sellers and one buyer. Each product independently has a good match value of $V = 100$ points with probability $q = 0.5$ and a bad match value of zero otherwise. I vary the cost of searching the second seller between sessions. The low-cost treatment has $s = 20$ and the high-cost treatment has $s = 40$. All other market rules are the same, and subjects participate in only one treatment.

Sellers bid for positions, learn their positions and the position payment, and then set prices. The buyer observes both prices and the top match before deciding whether to search the second seller. She can buy the second seller's product only after search and can return to the top seller for free. Bids and prices are restricted to whole points from 0 to 100. The continuous-price equilibrium provides the theoretical benchmark for this integer-grid implementation.

Roles are assigned at the beginning and remain fixed. In each round, the computer randomly and anonymously rematches subjects into groups of two sellers and one buyer and independently draws new match values. Subjects receive written instructions and answer three comprehension questions before completing two unpaid practice rounds and 30 incentivized rounds. At the end of each incentivized round, they receive feedback on the realized history of their market and their own payoff. Feedback concerns information revealed during play. The second match value is not disclosed when the buyer has not searched that product.

Every round begins with a 200-point balance, reset independently of earlier earnings. This balance covers possible losses from search and position payments. One of the 30 incentivized rounds is randomly selected for payment. The selected round's balance plus payoff is converted at 10 points per dollar and added to a \$10 show-up fee. The welfare measures exclude the initial balance and show-up payment and measure surplus generated within the market. Appendix F reproduces the instructions.

3.2 Laboratory Implementation

I conducted six computerized sessions at the University of Virginia's Veconlab, with three sessions per treatment. There were 42 subjects in the low-cost treatment and 45 in the high-cost treatment. Five sessions had 15 subjects and one low-cost session had 12. The low-cost treatment included 28 sellers and 14 buyers, while the high-cost treatment included

30 sellers and 15 buyers. With 30 incentivized rounds per subject, the sample contains 420 low-cost and 450 high-cost market observations.

Each market in each round is one observation for the outcome comparisons, which treat these observations as independent. Search regressions use session-clustered standard errors and compare them with buyer-clustered and unclustered standard errors.

3.3 Hypotheses

The hypotheses apply the continuous equilibrium to the experimental parameters. Numerical predictions average over equilibrium prices and match draws before their realization. Section 4.1 also compares observed outcomes with benchmarks conditional on the realized match draws. In both calculations, the buyer's decisions use only the information available at the time of choice. Appendix D gives the calculations.

Hypothesis 1 (Bidding and platform revenue). Raising the search cost from 20 to 40 points raises equilibrium bids and platform revenue from 20 to 40 points.

The larger search disadvantage of the second seller increases the revenue advantage of the top position by 20 points. Competition for that position transfers the increase to the platform.

Hypothesis 2 (Prices). Top prices rise and second prices fall as search cost increases, with price distributions given by (9) and (10).

At $s = 20$, the top and second price supports are $[70, 100]$ and $[30, 60]$, with means of 81.59 and 49.93 points. At $s = 40$, the supports are $[90, 100]$ and $[10, 20]$, with means of 93.86 and 18.96 points. The top price distribution first-order stochastically dominates the second price distribution in each treatment.

Hypothesis 3 (Search). Search follows the match-specific cutoffs in Lemma 1 and becomes less frequent as search cost increases.

After a bad top match, the buyer searches when the second price is at most 60 points at $s = 20$ and at most 20 points at $s = 40$. After a good top match, she searches when the price gap exceeds 40 and 80 points, respectively. At equilibrium prices, a buyer with a bad top match always searches. Search after a good top match occurs with probabilities 26.65% and 8.06%, so the overall search probabilities are 63.32% and 54.03%.

Hypothesis 4 (Revenues, profits, and welfare). Higher search cost raises the top seller's expected revenue and lowers the second seller's expected revenue, both sellers' expected profits, consumer surplus, and total surplus.

Expected top and second revenues change from 35 and 15 points at $s = 20$ to 45 and 5 points at $s = 40$. Each seller's expected profit falls from 15 to 5 points, so combined seller

profits fall from 30 to 10 points. Expected total surplus falls from 62.34 to 53.39 points and expected consumer surplus falls from 12.34 to 3.39 points. These are unconditional predictions. The realized-draw benchmarks used in the empirical comparison can differ because the observed frequencies of match pairs need not equal their ex ante probabilities.

4 Experimental Results

I first compare welfare outcomes with equilibrium, then examine the bidding, pricing, and search decisions behind these outcomes. I also compare revenues and profits at the two positions to distinguish the return from occupying the top position from the return after paying for it. The analysis uses all 30 incentivized rounds. Unless stated otherwise, reported means pool the 420 low-cost and 450 high-cost markets, giving each market equal weight.

4.1 Welfare

For each market, consumer surplus is the value of the purchased product minus its price and any search cost. Platform revenue is the losing bid paid by the top seller. Combined seller profits equal the two sellers' sales revenues less that payment. Equation (17) gives the accounting identity: prices and position payments redistribute surplus, while allocation and search determine its total. An search cost is counted even when the buyer exits or returns to the top seller.

The equilibrium comparison uses the same match draws as the experiment. For each market, I calculate the expected equilibrium outcome conditional on its realized pair (v_1, v_2), integrating over the equilibrium price distributions, and then average these conditional predictions. Buyers in this calculation retain the model's information structure: they do not learn v_2 before deciding whether to search. Conditioning adjusts the comparison for the finite sample of match draws. It does not give the benchmark buyer additional information. In particular, the equilibrium platform payment remains s in every market. The observed-draw adjustment can make the two positions' sample-average equilibrium profits differ even though their ex ante equilibrium profits are equal.

Figure 5 and Table 1 report the welfare outcomes. Within each treatment, observed outcomes are paired with their market-specific equilibrium benchmarks. Across treatments, I use Welch tests. The last column of the table tests whether the observed treatment change equals the predicted change, using the difference between observed and predicted outcomes in each market. Each market in each round is treated as an independent observation for these tests. All tests are two-sided and unadjusted for multiple comparisons.

Total surplus. Observed total surplus falls from 58.62 at $s = 20$ to 50.09 at $s = 40$, a decline of 8.53 points ($p = 0.013$). The draw-conditional equilibrium benchmarks are 64.03

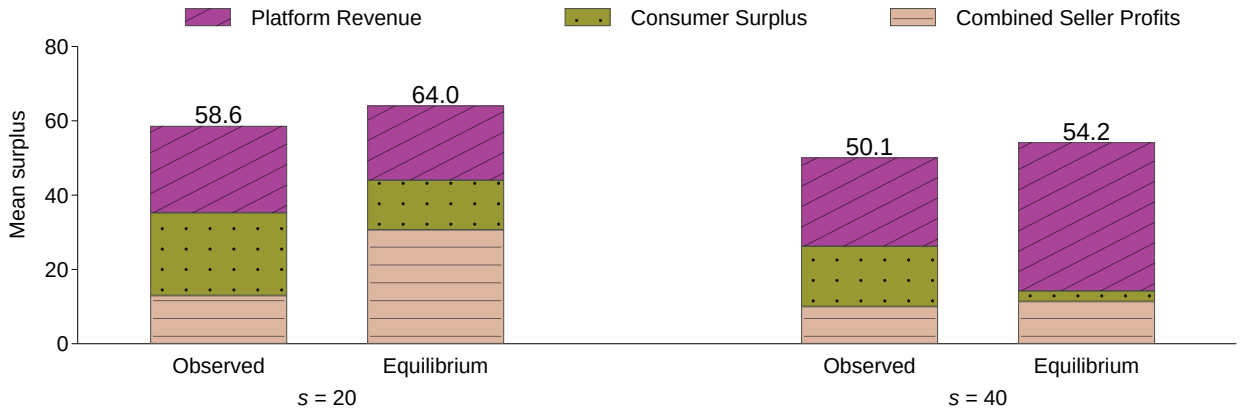


Figure 5: The level and distribution of surplus

Means in experimental points. Each bar is total surplus, divided into platform revenue, consumer surplus, and combined seller profits. Equilibrium outcomes condition on the observed match draws and integrate over equilibrium prices. Values above the bars report total surplus. Table 1 reports each component and comparison. The figure displays means without confidence intervals.

Table 1: Welfare outcomes and equilibrium benchmarks

	$s = 20$		$s = 40$		Treatment change $(40 - 20)$		
Outcome	Observed	Theory	Observed	Theory	Observed	Predicted	p_{pred}
Total surplus	58.62***	64.03	50.09***	54.23	-8.53	-9.80	0.513
Platform revenue	23.35***	20.00	23.83***	40.00	0.48	20.00	< 0.001
Consumer surplus	22.29***	13.34	16.23***	2.84	-6.06	-10.50	0.013
Combined seller profits	12.98***	30.69	10.03	11.39	-2.95	-19.29	< 0.001

Means are points per market over all 30 rounds ($N = 420$ at $s = 20$ and $N = 450$ at $s = 40$). Stars on Observed test equality with the corresponding Theory mean using paired market differences: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Changes are high minus low. The p_{pred} column tests equality of the observed and predicted changes using a Welch test on market-level observed-minus-theory outcomes. All tests are two-sided and treat markets as independent. Theory conditions on each market's realized match values and averages over equilibrium pricing and buyer decisions. Seller profits subtract the payment for prominence.

and 54.23, a decline of 9.80 points. Thus, the observed change has the predicted direction, while total surplus is below equilibrium in both treatments. The observed-minus-equilibrium gaps are -5.41 at the low cost and -4.15 at the high cost (paired $p < 0.001$ and $p = 0.004$).

The accounting in (17) clarifies these gaps. The mean allocated match value is 66.67 at the low cost and 60.22 at the high cost, compared with benchmarks of 76.90 and 75.11. Observed search expenditure is also lower: 8.05 versus 12.87, and 10.13 versus 20.88. Fewer resources are spent on search, but the reduction in realized allocation value is larger. This accounting shows how the two components of total surplus differ from their equilibrium values. It does not separately attribute these differences to prices and buyer choices.

Result 1. Total surplus is below the equilibrium benchmark in both treatments and decreases as search cost rises.

Platform revenue. The predicted 20-point increase in auction revenue does not appear in the pooled means. Platform revenue is 23.35 at $s = 20$ and 23.83 at $s = 40$, an increase of 0.48 points ($p = 0.718$). The 95% Welch interval for this increase is $[-2.11, 3.07]$. This estimate is small relative to the predicted increase of 20 points, although nonrejection of equal observed means does not establish equality. Revenue exceeds the low-cost equilibrium benchmark by 3.35 points and falls below the high-cost benchmark by 16.17 points (both market-level $p < 0.001$). The relevant auction statistic is the losing bid, which determines the payment, rather than the average submitted bid.

Result 2. Platform revenue changes little as search cost rises, despite the predicted increase of 20 points. It exceeds the equilibrium benchmark at the low search cost and falls below it at the high search cost.

Consumer surplus. Consumers receive more surplus than predicted in both treatments. Observed consumer surplus is 22.29 at $s = 20$, against a benchmark of 13.34, and 16.23 at $s = 40$, against 2.84 (both paired $p < 0.001$). It falls by 6.06 points across treatments ($p = 0.002$), compared with a predicted decline of 10.50 points.

The proportional comparison is sharper: observed consumer surplus falls by 27.2%, whereas its draw-conditional equilibrium benchmark falls by 78.7%. These percentages are changes in treatment means, not averages of individual percentage changes. A delta-method comparison, retaining the covariance between observed and equilibrium surplus within each treatment, gives $p < 0.001$ for the 51.5-percentage-point difference. The higher level of consumer surplus is compatible with lower total surplus: consumers make lower payments and incur less search expenditure, while some valuable purchases fail to occur.

Result 3. Consumer surplus exceeds the equilibrium benchmark in both treatments. It decreases with search cost, but the proportional decline is smaller than predicted.

Combined seller profits. The two sellers together earn 12.98 after the position payment at $s = 20$, well below the equilibrium benchmark of 30.69 (paired $p < 0.001$). At $s = 40$, their mean profit is 10.03 against a benchmark of 11.39. The 1.36-point shortfall is comparatively small and is not statistically detected ($p = 0.358$). The across-treatment difference is also imprecisely estimated ($p = 0.233$). Sellers retain 22.1% and 20.0% of observed total surplus, respectively. These shares are ratios of treatment means, not averages of market-specific ratios.

The proximity to equilibrium profit at the high search cost does not imply that seller behavior or revenues are close to equilibrium. As Table 4 shows, combined sales revenue is 36.33 rather than 50.69 at the low cost and 33.86 rather than 51.39 at the high cost. Revenue at each position is below its draw-conditional benchmark at both costs (all paired market-level $p < 0.001$). The profit identity explains the different profit gaps. At $s = 20$, a revenue shortfall of 14.36 and an auction payment 3.35 above equilibrium together produce the 17.71-point profit shortfall. At $s = 40$, the revenue shortfall of 17.53 is largely offset by an auction payment 16.17 below equilibrium, leaving a profit shortfall of 1.36. This is an exact accounting relation, not a decomposition into independently identified behavioral effects.

Result 4. The sellers jointly retain about one fifth of total surplus after paying for prominence. Their combined profit is below equilibrium at the low search cost and is much closer to the benchmark at the high search cost.

4.2 Bidding, Pricing, and Search

Bidding and the platform payment. Figure 6 and Table 2 distinguish pooled, winning, and losing bids. The mean submitted bid rises from 34.03 to 38.79, an increase of 4.76 points. At the market level, this increase differs from zero and is smaller than the predicted increase of 20 (both $p < 0.001$). For this test, the pooled-bid observation is the average of the two bids within a market, so the two bids are not treated as separate observations. The pooled mean is above the equilibrium bid of 20 in the low-cost treatment ($p < 0.001$). Its difference from 40 in the high-cost treatment is not detected ($p = 0.221$).

Winning bids rise from 44.71 to 53.76, an increase of 9.05 points rather than the predicted 20. Losing bids remain near 24. Since the top seller pays the losing bid, the increase in winning bids does not itself increase platform revenue. The high-cost pooled mean also conceals substantial differences between winning and losing bids: a mean near 40 need not imply that individual bids concentrate near the equilibrium bid.

Result 5. Mean submitted bids rise by 4.76 points as search cost increases, well short of the predicted increase of 20 points. Winning bids rise, while losing bids change little.

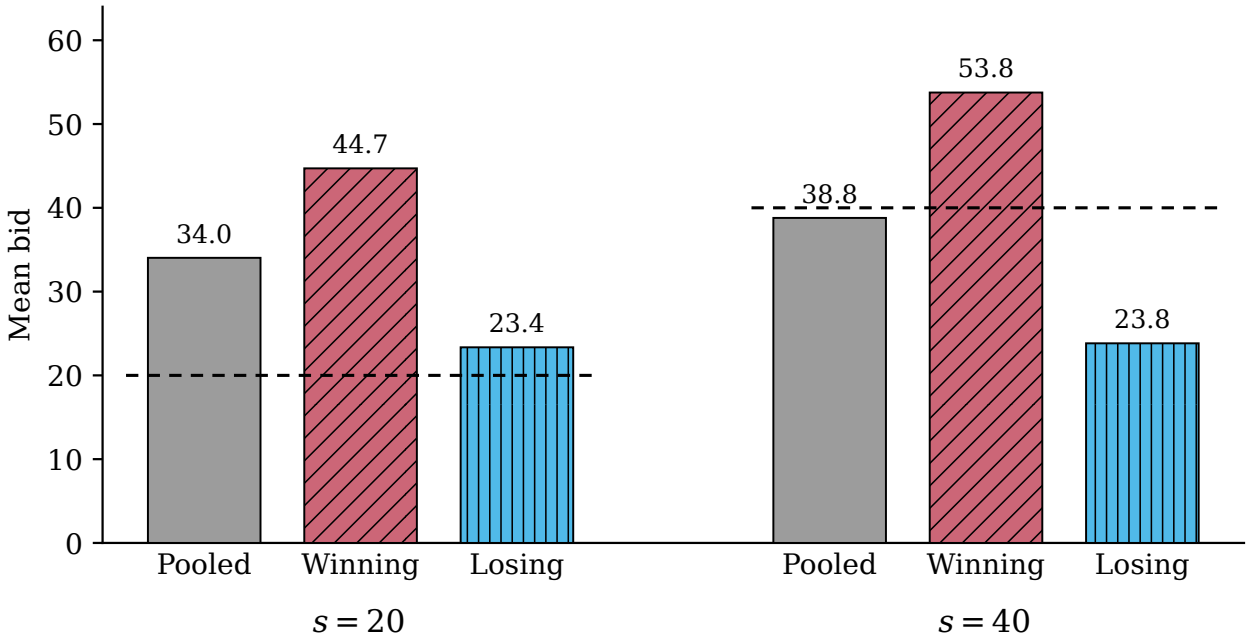


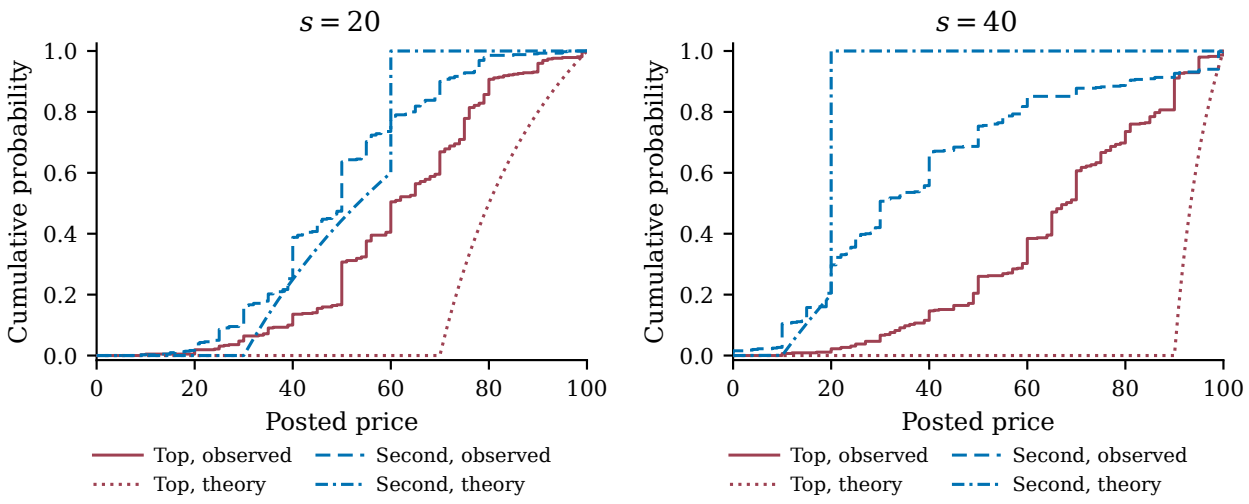
Figure 6: Submitted bids and payments for prominence

Means over all 30 rounds. The pooled bid is the mean of both bids in a market. The winning bid determines the top position, while the losing bid determines the payment to the platform. Horizontal reference lines show the equilibrium bids of 20 and 40 points.

Table 2: Bids and posted prices

	$s = 20$		$s = 40$		Treatment change $(40 - 20)$		
Outcome	Observed	Theory	Observed	Theory	Observed	Predicted	p_{pred}
Top price	61.83***	81.59	66.07***	93.86	4.24	12.27	< 0.001
Second price	48.90	49.93	39.11***	18.96	-9.79	-30.97	< 0.001
Pooled bid	34.03***	20.00	38.79	40.00	4.76	20.00	< 0.001
Winning bid	44.71***	20.00	53.76***	40.00	9.05	20.00	< 0.001
Losing bid	23.35***	20.00	23.83***	40.00	0.48	20.00	< 0.001

Means are points per market over all 30 rounds ($N = 420$ at $s = 20$ and $N = 450$ at $s = 40$). Stars on Observed test equality with the corresponding Theory mean using paired market differences: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Changes are high minus low. The p_{pred} column tests equality of the observed and predicted changes using a Welch test on market-level observed-minus-theory outcomes. All tests are two-sided and treat markets as independent. Theory prices are analytical equilibrium means and theory bids equal the search cost. The pooled bid is the mean of the two submitted bids in a market and contributes one observation. The losing bid equals the top seller's payment to the platform.



(a) Low search cost

(b) High search cost

Figure 7: Observed and equilibrium price distributions

All 30 rounds. Red and blue distinguish the top and second sellers. Solid and dashed curves show the observed CDFs, while dotted and dash-dot curves show their equilibrium CDFs. These distributions concern posted prices, whether or not a sale occurs.

Pricing. Figure 7 compares observed and equilibrium price distributions. Top sellers post substantially lower prices than predicted: their means are 61.83 and 66.07, compared with equilibrium means of 81.59 and 93.86 (one-sample market-level $p < 0.001$ at both costs). The second seller's mean is 48.90 against 49.93 at the low cost ($p = 0.196$), but 39.11 against 18.96 at the high cost ($p < 0.001$). These tests compare observed prices with the fixed analytical equilibrium means. The posted-price distributions also extend well beyond the equilibrium supports. Top-price standard deviations are 17.48 and 20.27, against predicted values of 8.39 and 2.80. Second-price standard deviations are 16.30 and 25.36, against 10.57 and 2.44. These dispersion comparisons are descriptive.

Top prices exceed second prices in 66.7% of low-cost markets and 81.6% of high-cost markets. The corresponding mean gaps are 12.93 and 26.96, below the predicted 31.65 and 74.90. These paired comparisons establish neither a uniform price ordering nor first-order stochastic dominance: the empirical cumulative distributions cross. More importantly, an average price gap does not determine search after a bad top match. In that case, the relevant comparison is the second price with $100 - 2s$, independently of the top price.

The second seller exceeds that bad-match search cutoff in 21.4% of all low-cost markets and 70.2% of all high-cost markets. Among buyers who actually receive a bad top match, the second price lies strictly above the cutoff in 46 of 216 low-cost observations and 143 of 216 high-cost observations. At those prices, stopping is optimal. The implications for sales and buyer surplus depend on the search choices discussed next.

Table 3: Search decisions relative to the optimal cutoff

Top match	s	Strict cases	Compliant	Under-search	Over-search	Ties
Good	20	199	90.5%	9	10	5
Good	40	233	89.7%	1	23	1
Bad	20	210	80.5%	32	9	6
Bad	40	195	64.6%	20	49	21

Compliance classifies the search decision at observed prices. Exact search ties are excluded from the strict cases and reported separately. Under-search means stopping when search has strictly positive net expected value. Over-search means searching when stopping is strictly preferable. Counts of errors are not conditional error rates. Purchase optimality is a separate decision.

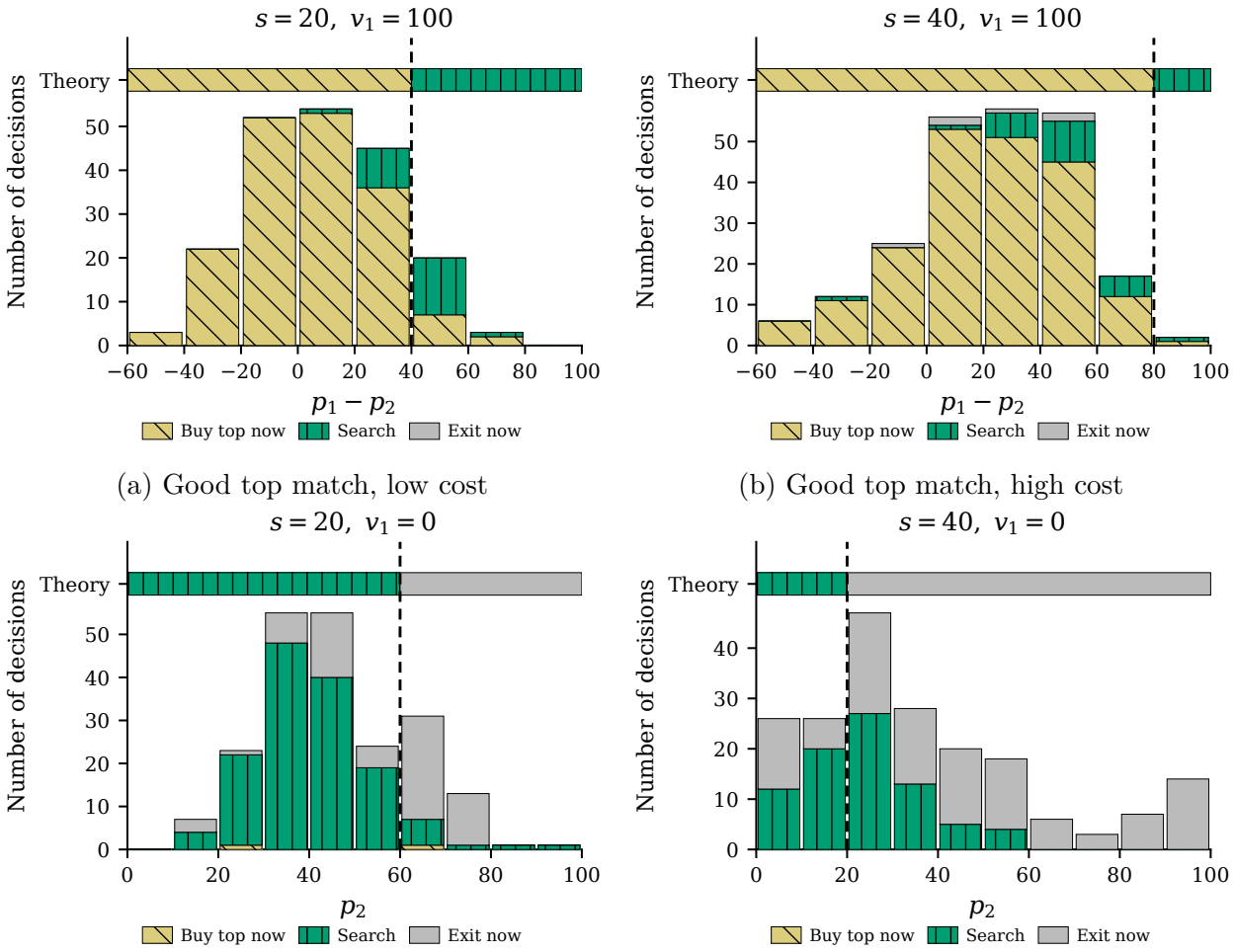
Result 6. Prices at both positions are more dispersed than predicted, with observed prices extending beyond the equilibrium supports. Top sellers charge higher prices on average than second sellers, but lower prices than predicted. Second sellers charge more than predicted at the high search cost.

Search decisions. I evaluate search choices at the prices subjects actually observe. With a good top match, search is strictly optimal when $p_1 - p_2 > 2s$. With a bad top match, it is strictly optimal when $p_2 < 100 - 2s$. Figure 8 displays decisions around these cutoffs. Table 3 excludes exact search ties when calculating compliance and reports their counts separately. This avoids classifying an indifferent buyer's search choice as a strict error. The equilibrium tie convention remains relevant to the theoretical benchmark, but the empirical accuracy measure does not require subjects to follow that convention.

After a good top match, 90.5% of low-cost and 89.7% of high-cost decisions satisfy the strict search rule. These high rates partly reflect the prices buyers face. In the high-cost treatment, 231 of the 233 strict cases call for stopping, leaving only two cases in which search is strictly preferable. Overall compliance therefore provides limited evidence about behavior on both sides of that cutoff. After a bad top match, compliance is lower: 80.5% at $s = 20$ and 64.6% at $s = 40$.

Errors occur in both directions. Low-cost buyers with a bad top match fail to search in 32 of the 164 cases in which search is strictly optimal, and search in 9 of the 46 cases in which stopping is optimal. At the high cost, the corresponding counts are 20 of 52 and 49 of 143. The conditional under-search rates are therefore 19.5% and 38.5%, and the over-search rates are 19.6% and 34.3%. Under-search can prevent a valuable second purchase. Over-search incurs an search cost when it is not justified by the buyer's expected private gain. Search accuracy is distinct from purchase accuracy after the available products have been revealed.

Purchase choices are much closer to the payoff-maximizing choice within the buyer's chosen information set: there are four purchase errors among 420 low-cost markets and six among 450 high-cost markets. The principal departures therefore concern how sellers bid and



(a) Good top match, low cost

(b) Good top match, high cost

(c) Bad top match, low cost

(d) Bad top match, high cost

Figure 8: Search decisions at observed prices

All 30 rounds. Bar heights show the frequency of observed prices or price differences. Segments distinguish buying from the top seller, searching the second seller, and exit at the first decision. The price difference governs search after a good top match. The second price governs it after a bad top match. The strip marked Theory indicates the optimal action in each price region. Cutoff ties are shown in the figures but excluded from the strict-compliance rates in Table 3.

price and whether buyers search the second product, rather than which searched product buyers choose.

Result 7. Most search decisions satisfy the optimal cutoff after a good top match. Compliance is lower after a bad top match, particularly at the high search cost, with both under-search and over-search.

Pricing and search jointly explain how consumers can receive more surplus while sellers earn less revenue. Lower top prices raise the surplus from buying the top product. After a good top match, they also reduce the price gap and can make stopping optimal, saving the buyer a further search cost. The same lower prices reduce the top seller's revenue per sale.

After a bad top match, the second price determines whether further search is worthwhile. Prices above the cutoff discourage search even when the second product would turn out to be a good match. This is an optimal response to the posted price, rather than a buyer mistake. Under-search at prices below the cutoff can prevent additional valuable purchases. Over-search, in contrast, incurs a cost when the expected private gain is too small. These different responses must be distinguished when interpreting the lower sales revenues and total surplus.

The payment for prominence then determines how much of the sellers' revenue remains as profit. At the low search cost, lower revenue and an above-equilibrium payment both reduce seller profits. At the high search cost, the payment shortfall offsets much of the revenue shortfall. Similar profits can therefore conceal very different pricing and bidding behavior.

4.3 Returns to prominence

Combined profits conceal how revenues and payments differ between positions. Table 4 separates the two sellers' revenues and profits. The top seller earns more revenue on average in both treatments, but it is also the only seller that pays the platform.

The observed revenue advantage of the top position is 19.01 points at the low search cost and 27.91 points at the high search cost. The corresponding mean payments are 23.35 and 23.83 points. Thus, at the low cost, the payment exceeds the observed revenue advantage. Average top profit is 4.32 points, compared with 8.66 for the second seller. At the high cost, top profit is 7.06 points and second profit is 2.97 points. Average revenue is higher in the top position, but average profit need not be higher after paying for it.

These comparisons satisfy the market-by-market identity

$$\pi_1 - \pi_2 = (\Pi_1 - \Pi_2) - b^{(2)}. \tag{18}$$

The distinction between a bid and a payment is also important. Mean submitted bids of 34.03 and 38.79 points exceed the observed revenue advantages of 19.01 and 27.91. The

Table 4: Revenues, profits, and returns to prominence

	$s = 20$		$s = 40$		Treatment change $(40 - 20)$		
Outcome	Observed	Theory	Observed	Theory	Observed	Predicted	p_{pred}
Top revenue, Π_1	27.67***	34.17	30.89***	46.73	3.22	12.57	< 0.001
Second revenue, Π_2	8.66***	16.52	2.97***	4.66	-5.69	-11.86	< 0.001
Combined revenue	36.33***	50.69	33.86***	51.39	-2.47	0.71	0.030
Revenue advantage, $\Pi_1 - \Pi_2$	19.01	17.64	27.91***	42.07	8.91	24.43	< 0.001
Payment for prominence	23.35***	20.00	23.83***	40.00	0.48	20.00	< 0.001
Top profit, π_1	4.32***	14.17	7.06	6.73	2.74	-7.43	< 0.001
Second profit, π_2	8.66***	16.52	2.97***	4.66	-5.69	-11.86	< 0.001
Profit difference, $\pi_1 - \pi_2$	-4.34	-2.36	4.09	2.07	8.43	4.43	0.081

Means are points per market over all 30 rounds ($N = 420$ at $s = 20$ and $N = 450$ at $s = 40$). Stars on Observed test equality with the corresponding Theory mean using paired market differences: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Changes are high minus low. The p_{pred} column tests equality of the observed and predicted changes using a Welch test on market-level observed-minus-theory outcomes. All tests are two-sided and treat markets as independent. Theory conditions on realized match values. Revenue excludes the prominence payment and top profit subtracts it. Position differences describe auction-assigned positions and do not identify a causal effect of moving a seller between positions.

second-price auction does not charge the winning seller its own bid, so this comparison alone does not measure a realized loss from bidding.

Nor is the observed revenue advantage a direct estimate of any seller's willingness to pay. Sellers are selected into positions by their bids, and they bid before observing the subsequent prices and match values. Evaluating an individual bid requires that seller's expected continuation revenues at the time of bidding. The position-specific averages describe the returns realized in these markets. They do not establish what the same seller would have earned in the other position.

5 Individual decisions and adjustment over time

The aggregate findings describe the market outcomes and the behavior behind them. This section examines individual bidding, the relationship between bidding and pricing, and buyers' search decisions over repeated markets.

5.1 Individual bidding and adjustment

5.2 Bids, payments, and prices

5.3 Search decisions and adjustment

The cutoff comparisons describe whether buyers search when search is optimal at the observed prices. I now examine how the probability of search varies with each price, conditional on

the top match, and whether these relationships change over the session. This separates the response to the second price after a bad top match from the response to the price gap after a good top match.

Let G indicate a good top match and let r denote the round. For each search cost s, I estimate a separate logit model using all 30 incentivized rounds:

$$\Pr(x = 1 \mid G, p_1, p_2, r, s) = \Lambda(\eta_s),$$
$$\eta_s = \alpha_s + \gamma_s G + \beta_{1s} p_1 + \beta_{2s} p_2 + \delta_{1s} G p_1 + \delta_{2s} G p_2 + \rho_s r. \tag{19}$$

Here Λ is the logistic distribution function. Both price slopes can differ by top match, and every coefficient can differ by treatment. The round trend is common to the two match states within a treatment. The model uses information available when the buyer chooses whether to search and excludes the hidden second match and later purchase outcome. The estimates describe conditional price associations rather than causal price effects.

The theory suggests two restrictions on the index:

$$\begin{aligned} H_B: &\quad \beta_{1s} = 0, \\ H_G: &\quad \beta_{1s} + \delta_{1s} + \beta_{2s} + \delta_{2s} = 0. \end{aligned} \tag{20}$$

After a bad top match, the top price should not affect search. After a good top match, dependence on the price gap requires the two price coefficients to have equal magnitudes and opposite signs. These restrictions concern the specified index and do not impose the theory's deterministic cutoffs.

Table 5 reports average marginal effects with Session standard errors. Each state is evaluated at the same observed price and round vectors within its treatment.[2] The sample contains 420 low-cost and 450 high-cost decisions.

After a bad top match, a higher second price is associated with substantially less search. The average derivatives are -12.57 and -9.06 percentage points per 10 price points at the low and high search costs. The top-price derivatives are much smaller, at 1.99 and 2.08 percentage points. The direct tests of H_B do not reject under Session, Buyer, or Market inference.

After a good top match, a higher top price is associated with more search and a higher second price with less search. The derivatives are 5.56 and -9.23 at the low cost, compared with 3.09 and -3.54 at the high cost. Raising both prices equally while keeping their gap fixed therefore has an implied average derivative of -3.67 percentage points per 10 price points at

[2]Session denotes clustering by session, Buyer denotes clustering by buyer, and Market denotes heteroskedasticity-robust standard errors without clustering. There are only three sessions in each treatment, so the nominal Session Wald p-values are unreliable as calibrated measures of sampling uncertainty and are sensitive to the variance estimate. I report the comparison in Appendix E.4. There are 14 and 15 buyer clusters, respectively. The Session covariance has rank at most two, so tests imposing four price-slope restrictions are not reported.

Table 5: Search and prices: logit

Top match	Price	$s = 20$	$s = 40$
Bad	p_1	1.99 (1.98)	2.08 (1.71)
Bad	p_2	-12.57 (1.04)	-9.06 (1.38)
Good	p_1	5.56 (0.66)	3.09 (0.78)
Good	p_2	-9.23 (1.66)	-3.54 (0.85)
Index restrictions, nominal p-values			
Bad match, H_B		0.2696	0.2390
Good match, H_G		0.0015	5.63×10^{-6}
Observations		420	450

Average marginal effects in percentage points per 10 price points, with Session standard errors in parentheses. Each state is evaluated at every observed price and round vector within the treatment, setting G to zero or one. Effects are scaled local derivatives, not exact probability changes from a finite 10-point price increase. Separate treatment fits include a good-match indicator, both prices, both interactions with the good-match indicator, and a round trend. Restriction tests use the index coefficients and their full covariance, rather than separate tests of the marginal effects.

the low cost and -0.45 at the high cost. The latter point estimate is economically much closer to the gap-only restriction.

The Session logit tests nevertheless reject H_G in both treatments, with nominal p-values of 0.0015 and 5.63×10^{-6}. The corresponding Buyer values are 0.0889 and 0.5445, and the Market values are 0.0771 and 0.7204. Thus the signs of the relevant price responses agree with the search incentives, but the evidence against dependence on the price gap alone is sensitive to how uncertainty is estimated. A failure to reject under an alternative variance estimate does not establish the restriction.

Appendix E.1 replaces the logit link with probit. The average price derivatives remain similar, although the high-cost good-match restriction is no longer rejected with Session standard errors. Appendix E.2 allows nonlinear responses to the second price after a bad top match and to the price gap after a good top match. The remaining top-price and common-price associations are similar in magnitude to those in the baseline, with the low-cost bad-match top-price derivative becoming smaller.

To examine adjustment, I also allow every index coefficient to differ between the first 15 and last 15 rounds within each treatment. Both fitted rules are evaluated on the same reference prices, so the comparison does not combine a change in the response function with a change in the price distribution. At the high search cost, the bad-match top-price derivative falls from 4.94 to 0.11 percentage points, but the direct difference has a Session p-value of 0.4180. The low-cost counterpart changes from 1.68 to 1.99, with $p = 0.8753$.

These estimates do not establish that top-price dependence disappears with experience. The complete comparisons in Appendix E.3 describe adjustment over the observed rounds, without identifying a learning process or long-run convergence.

6 A quantal-response model

7 Conclusion

This paper studies a market in which sellers compete for prominence before setting prices and consumers decide how far to search. The model links the value of the prominent position to the pricing and search decisions that follow its allocation. A laboratory experiment varies the cost of searching the second seller's product and observes how surplus is divided among consumers, sellers, and the platform.

The welfare results have three main features. Platform auction revenue changes little across search costs, reflecting the weak response of losing bids. Consumer surplus exceeds its equilibrium benchmark at both costs and falls proportionally less than predicted as the cost increases. Sellers jointly retain only about one fifth of total surplus after paying for prominence. Lower top prices benefit buyers who purchase from the top seller but reduce seller revenue per sale. Second prices above the search cutoff can discourage further search after a bad initial match and lead to lost sales. Combined sales revenue is below equilibrium in both treatments, and position payments take a substantial share of that revenue. At the high search cost, payments below equilibrium offset much of the revenue shortfall, explaining why profits are closer to the benchmark there.

The experiment separates the search-cost advantage of prominence from information that rankings may convey in the field. It also shows why bids and product-market outcomes should be studied together. Auction payments depend on sellers' valuations of prominence, while those valuations depend on anticipated pricing and consumer search. An auction model that takes position values as given can describe payment incentives, but cannot alone explain how a change in consumer search friction changes those values. Conversely, a model that assigns prominence without payment misses the transfer of seller revenue to the platform.

These findings suggest several directions for further work. One concerns platforms' incentives to change search costs. In the experiment, a lower cost improves consumer and total surplus without a statistically detectable decline in auction revenue. This comparison does not establish an optimal search cost for a platform: investment in search technology may be costly, and transaction commissions, participation, and other revenue sources may also respond. An endogenous platform decision would need to incorporate these margins rather than equate platform profit with auction revenue alone.

A second question is how prominence should be allocated. Comparing the present auction with random allocation or a first-price auction would show how outcomes depend on the allocation and payment mechanisms. With more sellers and several degrees of prominence, the allocation could also affect which products enter consumers' consideration sets. Seller differences in quality, costs, or match distributions would then matter for whether willingness to pay selects the socially preferred search order. The two identical sellers in this paper isolate the value of a search-cost advantage. They do not settle that broader allocation question.

A third question concerns the product information that sellers can display to consumers without further search. Here, prices are visible but match values must be learned through search. Other information structures may change how sellers direct attention and how buyers choose which products to search. Studying price visibility alongside product characteristics, reviews, and platform disclosure rules would help connect competition for prominence to the design of the search environment itself.

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Appendix A Derivation of buyer search and demand

A.1 Choices at indifference

The buyer behavior in Lemma 1 uses the following choices when several actions are optimal.

A1. With $v_1 = V$, the buyer stops when indifferent between searching the second seller and stopping.

A2. With $v_1 = 0$, the buyer searches the second seller when indifferent between searching and stopping.

A3. Among available choices yielding the same highest purchase surplus, the buyer prefers purchase from the top seller to purchase from the second seller, and purchase from the second seller to exit.

These rules select optimal buyer actions at equality and support the pricing equilibrium in Theorem 1. Figure 9 gives the complete buyer decision tree conditional on the observed prices and top match.

A.2 Proof of Lemma 1

At any $p_1, p_2 \geq 0$, stopping gives $\max\{0, v_1 - p_1\}$. The expected gain from search before paying its cost is

$$q \max\{0, v_1 - p_1, V - p_2\} + (1 - q) \max\{0, v_1 - p_1, -p_2\} - \max\{0, v_1 - p_1\}. \tag{21}$$

The buyer searches when this gain exceeds s and stops when it is smaller. A1 and A2 resolve equality. Once search is complete, the search cost is sunk, and A3 selects a surplus-maximizing terminal action.

For $0 \leq p_1, p_2 \leq V$ and $v_1 = 0$, the gain is $q(V - p_2)$, so A2 gives search exactly when $p_2 \leq V - \frac{s}{q}$. For $v_1 = V$, the gain is $q \max\{p_1 - p_2, 0\}$, so A1 gives search exactly when $p_1 - p_2 > \frac{s}{q}$. These prove the two cutoff rules. A price $p_1 > V$ yields no sales. If $p_2 > V - \frac{s}{q}$, the gain from search is at most $q \max\{V - p_2, 0\} < s$, so seller 2 is never searched. □

A.3 Derivation of seller demand

For $0 < p_1 \leq V$ and $0 < p_2 \leq V - \frac{s}{q}$, the following paths give the demand functions in (6).

(v_1, v_2)	$p_1 - p_2 > \frac{s}{q}$	$p_1 - p_2 \leq \frac{s}{q}$
$(0, 0)$	Search, then exit	Search, then exit
$(0, V)$	Search, buy 2	Search, buy 2
$(V, 0)$	Search, return to 1	Buy 1 immediately
(V, V)	Search, buy 2	Buy 1 immediately

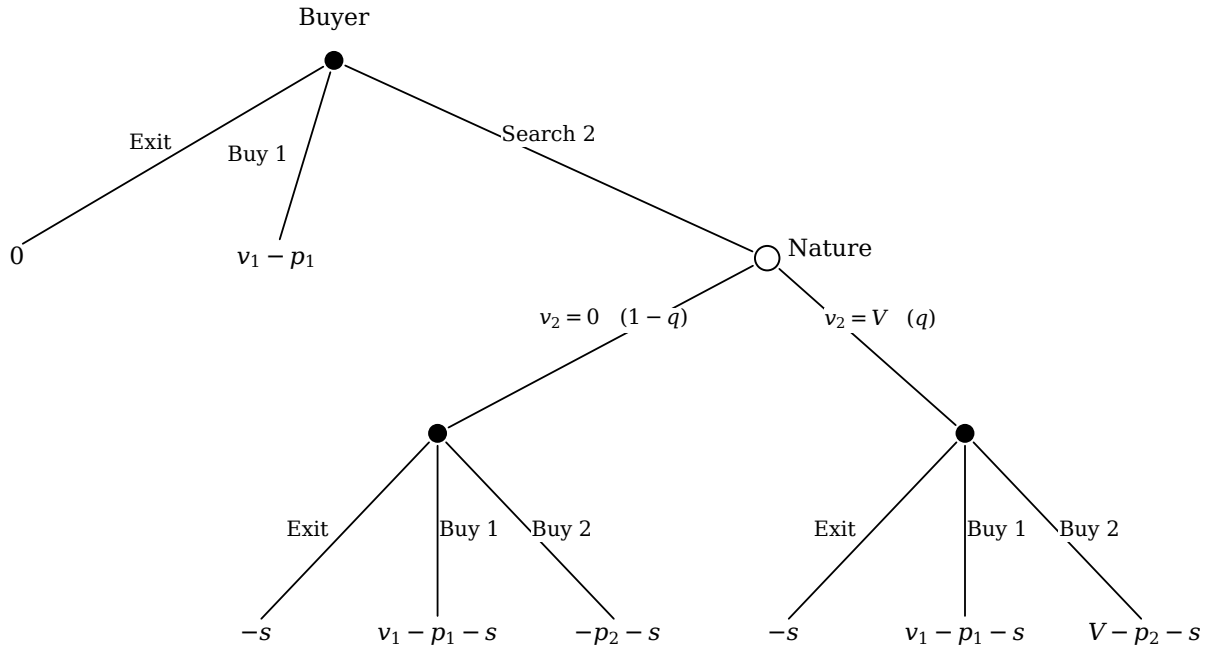


Figure 9: The buyer's complete decision tree

The buyer observes both prices and the top match before deciding whether to search the second seller. Search reveals the second match and costs s.

Appendix B Pricing equilibrium and price moments

B.1 Proof of Proposition 1

Each seller can guarantee strictly positive revenue. Seller 1 can charge any sufficiently small $0 < p_1 < \frac{s}{q}$ and sell whenever $v_1 = V$. Seller 2 can charge any $0 < p_2 < V - \frac{s}{q}$ and sell whenever $(v_1, v_2) = (0, V)$. Thus a pure equilibrium cannot involve a zero price, $p_1 > V$, or $p_2 > V - \frac{s}{q}$.

Consider the remaining domain $0 < p_1 \leq V$, $0 < p_2 \leq V - \frac{s}{q}$. If $p_1 - p_2 > \frac{s}{q}$, seller 2 can raise its price slightly while retaining demand q. If $p_1 - p_2 < \frac{s}{q}$, seller 1 can raise its price slightly while retaining demand q. At equality, seller 2 can choose a price strictly between $(1-q)p_2$ and p_2, increasing demand from $q(1-q)$ to q and strictly increasing revenue. Every candidate price pair therefore admits a profitable deviation. □

B.2 Proof of Theorem 1

The price bounds in Theorem 1 are

$$\begin{aligned} \underline{p}_1 &= (1-q)V + s, & \overline{p}_1 &= V, \\ \underline{p}_2 &= (1-q)\left(V - \frac{s}{q}\right), & \overline{p}_2 &= V - \frac{s}{q}. \end{aligned}$$

Both proposed CDFs are increasing on their continuous parts. The first rises from zero to one. The second starts at zero and approaches $1 - \frac{s}{qV}$ at its upper bound, with the remaining mass placed at that bound. Since $0 < s < qV$, both are valid distributions with strictly positive lower bounds.

Seller 1. For $0 < p_1 \leq V$, the expected revenue against F_2 is

$$p_1 \mathbb{E}_{F_2}[D_1(p_1, p_2)] = qp_1 \left[1 - q \Pr\left(p_2 < p_1 - \frac{s}{q}\right)\right]. \tag{22}$$

For $\underline{p}_1 \leq p_1 < V$, substitution of F_2 gives

$$p_1 \mathbb{E}_{F_2}[D_1(p_1, p_2)] = qp_1 \left[1 - \left(1 - \frac{(1-q)V + s}{p_1}\right)\right] = q[(1-q)V + s]. \tag{23}$$

At $p_1 = V$, a good-top-match buyer stops when $p_2 = V - \frac{s}{q}$ and searches at every lower second price. Including the mass at seller 2's upper bound therefore gives

$$V\left[q\frac{s}{qV} + q(1-q)\left(1 - \frac{s}{qV}\right)\right] = q[(1-q)V + s]. \tag{24}$$

A3 preserves seller 1's sales at zero purchase surplus. For $0 < p_1 < \underline{p}_1$, good-top-match buyers never search, so revenue is $qp_1 < q[(1-q)V + s]$. At $p_1 = 0$ or $p_1 > V$, revenue is zero. Hence every price in seller 1's support is a best response against all nonnegative pure-price deviations.

Seller 2. For $0 < p_2 \leq V - \frac{s}{q}$, bad-top-match buyers search. Good-top-match buyers search if $p_1 > p_2 + \frac{s}{q}$. Thus

$$p_2 \mathbb{E}_{F_1}[D_2(p_1, p_2)] = qp_2 \left[1 - qF_1\left(p_2 + \frac{s}{q}\right)\right]. \tag{25}$$

For $\underline{p}_2 \leq p_2 \leq \overline{p}_2$, substituting F_1 yields

$$p_2 \mathbb{E}_{F_1}[D_2(p_1, p_2)] = qp_2 \left[1 - \left(1 - \frac{(1-q)\left(V - \frac{s}{q}\right)}{p_2}\right)\right] = q(1-q)\left(V - \frac{s}{q}\right). \tag{26}$$

This calculation includes the upper bound: A2 ensures that bad-top-match buyers still search there. For $0 < p_2 < \underline{p}_2$, both top-match types search at every price in seller 1's support, giving revenue $qp_2 < q(1-q)(V - \frac{s}{q})$. A zero price gives zero revenue, and a price above $V - \frac{s}{q}$ attracts no search. Hence every price in seller 2's support is a best response against all nonnegative pure-price deviations.

The payoff from any mixed-price deviation averages the corresponding pure-price payoffs and cannot exceed these bounds. Both optimality conditions in Definition 1 therefore hold.□

B.3 Expected prices

$$\begin{aligned} \mathbb{E}[p_1] &= \frac{s}{q} + \frac{(1-q)\left(V - \frac{s}{q}\right)}{q} \log\left(\frac{1}{1-q}\right), \\ \mathbb{E}[p_2] &= \frac{(1-q)V + s}{q} \log\left(\frac{V}{(1-q)V + s}\right). \end{aligned} \tag{27}$$

Appendix C Bidding for prominence

C.1 Proof of Theorem 2

The two indifference calculations in (23) and (26) give the continuation revenues in (13). Their difference is

$$\begin{aligned}\mathbb{E}[\Pi_1] - \mathbb{E}[\Pi_2] &= q[(1-q)V + s] - q(1-q)\left(V - \frac{s}{q}\right) \\ &= qs + (1-q)s = s. \end{aligned} \tag{28}$$

Against an opponent bid b, winning rather than losing changes expected net profit by $s - b$. Bidding s wins when this difference is positive, loses when it is negative, and leaves the seller indifferent at equality. It is therefore weakly dominant conditional on the specified pricing continuation. With both sellers bidding s, the position payment is s and

$$\mathbb{E}[\pi_1] = q[(1-q)V + s] - s = (1-q)(qV - s) = \mathbb{E}[\pi_2].$$

The buyer is optimal after every price pair, and Theorem 1 supplies a pricing equilibrium after every auction outcome. These strategies and the symmetric bids therefore form a subgame-perfect equilibrium. □

Appendix D Calculation of revenues and welfare

D.1 Observed outcomes

For each market, the purchase indicator a and search indicator x determine consumer surplus and total surplus through (16) and (17). Seller revenue is its posted price if it makes the sale and zero otherwise. Platform revenue is the losing bid. Top profit subtracts that payment from top revenue, while second profit equals second revenue. The accounting identity is checked separately for every market before treatment means are calculated. Initial balances and show-up fees are excluded from all these measures.

To test an observed outcome against its conditional equilibrium benchmark, I first calculate their difference within each market. The within-treatment comparison is a one-sample test of whether the mean difference is zero. To test whether the treatment change matches the predicted change, I compare these differences across treatments with a Welch test. This preserves the pairing between an observed outcome and the benchmark based on the same match draws.

D.2 Benchmarks conditional on realized matches

The expectations below integrate over independent equilibrium prices, conditional on the indicated match draws. The price means are given in (27). For the calculation below, the lower price bounds are

$$\underline{p}_1 = (1-q)V + s, \qquad \underline{p}_2 = (1-q)\left(V - \frac{s}{q}\right).$$

After a good top match, search occurs when $p_1 - p_2 > s/q$. To calculate its probability, I integrate over the independent equilibrium price distributions in (9) and (10):

$$\begin{aligned}\Pr\left(p_1 - p_2 > \frac{s}{q}\right) &= \int_{\underline{p}_1}^{V} F_2\left(p_1 - \frac{s}{q}\right) dF_1(p_1) \\ &= \frac{\underline{p}_2}{q^2}\int_{\underline{p}_1}^{V} \frac{1 - \underline{p}_1/p_1}{\left(p_1 - \frac{s}{q}\right)^2}\, dp_1.\end{aligned}$$

The atom at the second seller's upper price bound contributes no search probability, since search at $p_2 = V - s/q$ would require $p_1 > V$. The top seller's distribution is atomless, so the value of the integrand at $p_1 = V$ does not affect the integral.

The logarithm comes from the reciprocal terms in the decomposition

$$\frac{1-\underline{p}_1/p_1}{\left(p_1-\frac{s}{q}\right)^2}=\frac{q^2\underline{p}_1}{s^2}\left(\frac{1}{p_1-\frac{s}{q}}-\frac{1}{p_1}\right)+\frac{1-q\underline{p}_1/s}{\left(p_1-\frac{s}{q}\right)^2}.$$

Integrating gives

$$\Pr\left(p_1-p_2>\frac{s}{q}\right)=\left[\frac{\underline{p}_1\underline{p}_2}{s^2}\log\left(\frac{p_1-\frac{s}{q}}{p_1}\right)-\frac{\underline{p}_2}{q^2}\frac{1-q\underline{p}_1/s}{p_1-\frac{s}{q}}\right]_{p_1=\underline{p}_1}^{p_1=V}.$$

Using $\underline{p}_1-s/q=\underline{p}_2=(1-q)(V-s/q)$, the difference between the two logarithms becomes

$$\log\left(\frac{V-\frac{s}{q}}{V}\frac{\underline{p}_1}{\underline{p}_1-\frac{s}{q}}\right)=\log\left(\frac{\underline{p}_1}{(1-q)V}\right).$$

The remaining term simplifies to $-\underline{p}_2/s$. Therefore,

$$\Pr\left(p_1-p_2>\frac{s}{q}\right)=\frac{\underline{p}_1\underline{p}_2}{s^2}\log\left(\frac{\underline{p}_1}{(1-q)V}\right)-\frac{\underline{p}_2}{s}. \tag{29}$$

When both matches are good, expected seller revenues are

$$\mathbb{E}[\Pi_1\mid V,V]=\mathbb{E}\left[p_1\cdot\mathbf{1}_{\{p_1-p_2\le\frac{s}{q}\}}\right]=\frac{(1-q)V+s-(1-q)\mathbb{E}[p_1]}{q}, \tag{30}$$

$$\mathbb{E}[\Pi_2\mid V,V]=\mathbb{E}\left[p_2\cdot\mathbf{1}_{\{p_1-p_2>\frac{s}{q}\}}\right]=\frac{(1-q)\left(V-\frac{s}{q}\right)-(1-q)\mathbb{E}[p_2]}{q}. \tag{31}$$

These identities follow by conditioning on the seller's own price and substituting the other seller's CDF. With $(v_1,v_2)=(0,V)$, only seller 2 makes a sale and earns p_2. With $(V,0)$, only seller 1 makes a sale and earns p_1. With $(0,0)$, neither makes a sale. A bad-top-match buyer always searches at equilibrium prices, whereas a good-top-match buyer searches with the probability in (29). Applying the surplus identities in (16) and (17) gives Table 6.

Table 6: Conditional equilibrium outcomes and observed match frequencies

s	(v_1, v_2)	Count	$\mathbb{E}[\Pi_1]$	$\mathbb{E}[\Pi_2]$	$\mathbb{E}[\text{CS}]$	$\mathbb{E}[\text{TS}]$
20	$(0, 0)$	97	0.00	0.00	-20.00	-20.00
20	$(0, 100)$	119	0.00	49.93	30.07	80.00
20	$(100, 0)$	105	81.59	0.00	13.08	94.67
20	$(100, 100)$	99	58.41	10.07	26.19	94.67
40	$(0, 0)$	112	0.00	0.00	-40.00	-40.00
40	$(0, 100)$	104	0.00	18.96	41.04	60.00
40	$(100, 0)$	113	93.86	0.00	2.91	96.77
40	$(100, 100)$	121	86.14	1.04	9.60	96.77

Expectations are conditional on each match pair. Counts sum to 420 and 450. Platform revenue equals s in every row, top profit is top revenue minus s, and second profit equals second revenue. Treatment benchmarks weight the rows by their observed frequencies. Values are rounded for display.

D.3 Unconditional predictions

Let ρ denote the probability of search after a good top match in (29). A buyer with a bad top match always searches at equilibrium prices. Averaging over the initial match therefore gives

$$\mathbb{E}[x] = (1 - q) + q\rho. \tag{32}$$

At equilibrium, a product with value V is purchased whenever at least one match is good. The probability is $1 - (1 - q)^2 = q(2 - q)$. Hence

$$\mathbb{E}[\text{TS}] = q(2 - q)V - s[(1 - q) + q\rho], \tag{33}$$

$$\mathbb{E}[\text{CS}] = \mathbb{E}[\text{TS}] - \mathbb{E}[\Pi_1] - \mathbb{E}[\Pi_2]. \tag{34}$$

These formulas yield the unconditional values in Section 3.3. The empirical benchmarks instead weight the conditional outcomes in Table 6 by the observed match frequencies. The buyer's search decision in either calculation depends only on the initial match and posted prices, not the second match before search.

Appendix E Additional empirical analyses

E.1 Probit and the search specification

The probit model replaces Λ in (19) with the standard normal distribution function and otherwise uses the same seven regressors and observations. Table 7 shows that the price derivatives closely resemble the logit estimates. Search falls with the second price after a bad top match. After a good top match, it rises with the top price and falls with the second price. These comparisons concern marginal effects because the index coefficients of logit and probit have different scales.

Table 7: Search and prices: probit

Top match	Price	$s = 20$	$s = 40$
Bad	p_1	2.16	2.03
		(1.66)	(1.59)
Bad	p_2	-12.06	-9.04
		(1.11)	(1.38)
Good	p_1	5.41	3.01
		(0.71)	(0.67)
Good	p_2	-8.92	-3.31
		(1.58)	(0.57)
Observations		420	450

Average marginal effects in percentage points per 10 price points, with Session standard errors in parentheses. Each state is evaluated at every observed price and round vector within the treatment, setting G to zero or one. Effects are scaled local derivatives, not exact probability changes from a finite 10-point price increase. Separate treatment fits include a good-match indicator, both prices, both interactions with the good-match indicator, and a round trend.

The point estimates do not imply identical restriction tests across links. In particular, the high-cost good-match restriction has a nominal Session p-value of 0.1457 under probit, compared with 5.63×10^{-6} under logit. Table 11 reports both restrictions for both links and all three variance estimates. The two fits retain all 870 observations. Estimating each treatment separately is equivalent to a pooled fit that interacts the treatment indicator with every term in the selected index.

E.2 Flexible responses to the relevant price

A linear price index can confound residual price dependence with an incorrectly specified response to the relevant price. I therefore add one nonlinear spline basis in each match state. In the bad state, the relevant argument is p_2 and a linear p_1 term remains unrestricted. In the good state, the relevant argument is $d = p_1 - p_2$ and a linear common-price term

$L = (p_1 + p_2)/2$ remains unrestricted. The treatment-specific index is

$$\eta_s = a_s + c_s G + \rho_s r + (1 - G)\{f_{Bs}(p_2) + \theta_{Bs} p_1\} + G\{f_{Gs}(d) + \theta_{Gs} L\}. \tag{35}$$

Each f is a restricted cubic spline with three knots and two basis terms. The fixed knots are $(19, 40, 70)$ for p_2 and $(-20, 20, 55)$ for d, based on the pooled support within each match state and applied identically to both treatments. They were not chosen by coefficient significance. This adds two parameters to the baseline, giving nine per treatment. The round trend remains common across match states.

The restrictions are $\theta_{Bs} = 0$ and $\theta_{Gs} = 0$. They test remaining linear top-price or common-price dependence, not all possible nonlinear departures from the theoretical restrictions. The good-match spline permits a smooth response to the gap without imposing an exactly flat segment for nonpositive gaps. A freely estimated negative-gap segment is unsuitable here because the buyer never searches in the 77 low-cost good-match observations with a nonpositive gap.

Table 8: Flexible price responses and remaining price dependence

		Partial effect		Index-null p-value		
s	Direction	Baseline	Flexible	Session	Buyer	Market
20	Bad, p_1	2.01	1.31 (1.65)	0.3969	0.2347	0.5050
20	Good, common level	-3.58	-3.49 (1.67)	0.0073	0.1490	0.1150
40	Bad, p_1	2.10	1.97 (1.73)	0.2610	0.2830	0.2336
40	Good, common level	-0.43	-0.49 (0.24)	5.45×10^{-6}	0.5059	0.6766

Partial effects are percentage points per 10 price points, averaged over the observed treatment and match-state sample. The Flexible column includes Session standard errors in parentheses. In the good state, both prices move equally so that the gap remains fixed. The p-values test a zero coefficient on bad-state p_1 or good-state common price level in the flexible index. They are direct coefficient tests, not Wald tests of the reported marginal effect. Both fits retain 420 observations at $s = 20$ and 450 at $s = 40$. The common within-state reference differs from the match-set reference in Table 5.

For the bad-match comparison, p_1 changes while p_2 remains fixed. For the good-match comparison, both prices change equally while the gap remains fixed. Consequently, the relevant spline argument does not change along either direction. The reported derivatives include the complete response in those directions. Baseline and flexible effects are averaged on identical observed treatment and match-state rows, with 216 bad-match observations in each treatment and 204 and 234 good-match observations. This reference isolates the change in functional form from a change in the evaluation sample.

The flexible estimates reduce the low-cost bad-match top-price derivative from 2.01 to 1.31 percentage points and leave its high-cost counterpart near 1.97. The good-match common-

price derivatives are -3.49 and -0.49. The Session tests reject the good-match common-price restriction in both treatments, whereas the Buyer and Market tests do not reject at 5%. The interpretation therefore remains sensitive to the variance estimate. The flexible fits converge without excluding observations.

E.3 Adjustment between the first and last 15 rounds

I interact an indicator for the last 15 rounds with every term in (19) and estimate the resulting model separately by treatment. Each half has its own intercept, good-match coefficient, four price terms, and round trend. Joint estimation retains the covariance between the first and last 15 rounds for the same buyer or session. There are 210 observations per half at the low cost and 225 per half at the high cost.

The reported marginal effects evaluate both half rules at the observed prices pooled across all 30 rounds within the treatment and match state. Round is fixed at 15.5 for both rules. This common midpoint uses the fitted linear round trends just outside each half's observed range. It avoids attributing a change in the price distribution to a change in the fitted price response. For each effect, the difference variance is

$$\begin{aligned}\operatorname{Var}(\widehat{m}_{\text{last}} - \widehat{m}_{\text{first}}) = \operatorname{Var}(\widehat{m}_{\text{last}}) + \operatorname{Var}(\widehat{m}_{\text{first}}) \\ - 2\operatorname{Cov}(\widehat{m}_{\text{last}}, \widehat{m}_{\text{first}}).\end{aligned} \tag{36}$$

All entries below use this full covariance calculation. A change from a significant coefficient in one half to an insignificant coefficient in the other is not a test of adjustment.

The high-cost bad-match top-price derivative is smaller in the last 15 rounds under both links. The direct changes are imprecise under each variance estimate, so they do not establish that this association disappears. At the low cost, the second-price derivative after a bad top match becomes more negative. Its logit change is -5.49 percentage points, with Buyer and Market p-values of 0.1052 and 0.1147. The probit change is -5.91, with corresponding values 0.0439 and 0.0856. The evidence for this particular change thus depends on link and variance choices. These are multiple descriptive comparisons over 30 rounds rather than estimates of a learning model.

Table 9: Adjustment across halves: logit

					p: last minus first		
s	Direction	First 15	Last 15	Difference	Session	Buyer	Market
20	Bad, p_1	1.68	1.99	0.31 (1.96)	0.8753	0.9141	0.9364
20	Bad, p_2	-10.33	-15.82	-5.49 (0.65)	2.65×10^{-17}	0.1052	0.1147
20	Good, p_1	6.23	4.86	-1.37 (1.03)	0.1822	0.4517	0.5672
20	Good, p_2	-8.80	-9.69	-0.89 (4.32)	0.8367	0.7153	0.8173
40	Bad, p_1	4.94	0.11	-4.83 (5.96)	0.4180	0.1875	0.1522
40	Bad, p_2	-7.42	-10.57	-3.15 (3.53)	0.3721	0.3455	0.2021
40	Good, p_1	1.73	3.39	1.65 (2.13)	0.4372	0.3491	0.3711
40	Good, p_2	-2.08	-4.25	-2.17 (3.58)	0.5445	0.3825	0.3632

Average marginal effects and differences are percentage points per 10 price points. Session standard errors of the differences are in parentheses. Both half rules use the same observed prices within each treatment and match state, with round fixed at 15.5. All seven index terms may differ across halves. Difference standard errors use the full covariance of the joint fit, including covariance between the two half estimates. Market denotes heteroskedasticity-robust inference without clustering.

Table 10: Adjustment across halves: probit

					p: last minus first		
s	Direction	First 15	Last 15	Difference	Session	Buyer	Market
20	Bad, p_1	1.67	2.31	0.64 (1.70)	0.7064	0.8080	0.8622
20	Bad, p_2	-9.78	-15.69	-5.91 (0.28)	1.45×10^{-95}	0.0439	0.0856
20	Good, p_1	6.05	4.72	-1.33 (1.18)	0.2587	0.4584	0.5622
20	Good, p_2	-8.53	-9.69	-1.17 (3.96)	0.7679	0.6172	0.7347
40	Bad, p_1	4.90	0.10	-4.80 (5.57)	0.3893	0.1628	0.1458
40	Bad, p_2	-7.42	-10.43	-3.01 (3.53)	0.3947	0.3536	0.2091
40	Good, p_1	1.55	3.61	2.06 (2.16)	0.3392	0.2350	0.2463
40	Good, p_2	-1.81	-4.28	-2.47 (3.42)	0.4701	0.2890	0.2534

Average marginal effects and differences are percentage points per 10 price points. Session standard errors of the differences are in parentheses. Both half rules use the same observed prices within each treatment and match state, with round fixed at 15.5. All seven index terms may differ across halves. Difference standard errors use the full covariance of the joint fit, including covariance between the two half estimates. Market denotes heteroskedasticity-robust inference without clustering.

E.4 Variance sensitivity and index coefficients

Table 11 makes the inferential comparison explicit. Session clusters by session, Buyer clusters by buyer, and Market uses heteroskedasticity-robust standard errors with each market treated as independent. The coefficient estimates are identical across these choices. Only the estimated covariance changes. The main discussion uses Session and reports the alternative tests because the good-match conclusion differs substantially across them.

Table 11: Search-rule restrictions under alternative variance estimates

Link	s	Null	Session	Buyer	Market
Logit	20	H_B	0.2696	0.1544	0.3170
Logit	20	H_G	0.0015	0.0889	0.0771
Logit	40	H_B	0.2390	0.2707	0.2098
Logit	40	H_G	5.63×10^{-6}	0.5445	0.7204
Probit	20	H_B	0.1467	0.0646	0.2806
Probit	20	H_G	0.0025	0.0640	0.0670
Probit	40	H_B	0.2116	0.2603	0.2106
Probit	40	H_G	0.1457	0.6415	0.7873

Two-sided nominal Wald p-values for the index restrictions in (20). Session and Buyer cluster by session and buyer. Market uses heteroskedasticity-robust standard errors without clustering. Changing the variance estimate leaves the fitted coefficients and probabilities unchanged. No multiplicity adjustment is applied.

Tables 12 and 13 provide all seven index coefficients and their standard errors under each variance estimate. Interaction coefficients alone are not the price responses after a good match. Those responses use $\beta_{1s} + \delta_{1s}$ and $\beta_{2s} + \delta_{2s}$, and their standard errors require the covariance between the relevant coefficients. The tests in Table 11 use those combinations directly.

Table 12: Search logit: index coefficients

	$s = 20$			$s = 40$		
Term	Session	Buyer	Market	Session	Buyer	Market
Constant, α_s	3.7778 (1.3355)	3.7778 (1.7500)	3.7778 (1.1373)	0.9655 (0.5842)	0.9655 (0.7811)	0.9655 (0.5934)
Good match, γ_s	-5.6709 (0.4016)	-5.6709 (1.9212)	-5.6709 (1.8532)	-3.8153 (0.6062)	-3.8153 (0.7349)	-3.8153 (1.0195)
p_1, β_{1s}	0.0113 (0.0103)	0.0113 (0.0080)	0.0113 (0.0113)	0.0105 (0.0089)	0.0105 (0.0096)	0.0105 (0.0084)
p_2, β_{2s}	-0.0716 (0.0042)	-0.0716 (0.0267)	-0.0716 (0.0159)	-0.0458 (0.0067)	-0.0458 (0.0109)	-0.0458 (0.0076)
Gp_1, δ_{1s}	0.0689 (0.0047)	0.0689 (0.0181)	0.0689 (0.0234)	0.0248 (0.0080)	0.0248 (0.0137)	0.0248 (0.0145)
Gp_2, δ_{2s}	-0.0616 (0.0077)	-0.0616 (0.0337)	-0.0616 (0.0351)	0.0054 (0.0092)	0.0054 (0.0141)	0.0054 (0.0156)
Round, ρ_s	-0.0116 (0.0103)	-0.0116 (0.0138)	-0.0116 (0.0178)	-0.0281 (0.0350)	-0.0281 (0.0222)	-0.0281 (0.0156)
Observations	420	420	420	450	450	450

Coefficients from the seven-term index in (19), with standard errors in parentheses. Prices enter in their original point units. Session and Buyer cluster by session and buyer. Market uses heteroskedasticity-robust standard errors without clustering. All 30 rounds are retained.

Table 13: Search probit: index coefficients

	$s = 20$			$s = 40$		
Term	Session	Buyer	Market	Session	Buyer	Market
Constant, α_s	2.0494 (0.6584)	2.0494 (0.8626)	2.0494 (0.5843)	0.5918 (0.3646)	0.5918 (0.4614)	0.5918 (0.3501)
Good match, γ_s	-3.1604 (0.1054)	-3.1604 (0.9919)	-3.1604 (0.9731)	-2.1779 (0.4139)	-2.1779 (0.4116)	-2.1779 (0.5416)
p_1, β_{1s}	0.0071 (0.0049)	0.0071 (0.0038)	0.0071 (0.0066)	0.0063 (0.0050)	0.0063 (0.0056)	0.0063 (0.0050)
p_2, β_{2s}	-0.0395 (0.0023)	-0.0395 (0.0133)	-0.0395 (0.0083)	-0.0279 (0.0041)	-0.0279 (0.0061)	-0.0279 (0.0043)
Gp_1, δ_{1s}	0.0369 (0.0022)	0.0369 (0.0092)	0.0369 (0.0125)	0.0121 (0.0046)	0.0121 (0.0078)	0.0121 (0.0081)
Gp_2, δ_{2s}	-0.0331 (0.0061)	-0.0331 (0.0171)	-0.0331 (0.0173)	0.0077 (0.0056)	0.0077 (0.0075)	0.0077 (0.0082)
Round, ρ_s	-0.0061 (0.0068)	-0.0061 (0.0081)	-0.0061 (0.0100)	-0.0169 (0.0193)	-0.0169 (0.0122)	-0.0169 (0.0088)
Observations	420	420	420	450	450	450

Coefficients from the seven-term index in (19), with standard errors in parentheses. Prices enter in their original point units. Session and Buyer cluster by session and buyer. Market uses heteroskedasticity-robust standard errors without clustering. All 30 rounds are retained.

Appendix F Experimental instructions[3]

Welcome! Thanks for participating in this experiment on economic decision-making. Your participation is completely voluntary. Please remain seated and do not use your cellphone. You will interact anonymously through the computers. Please do not talk or look at others' screen.

At the end of the experiment, we will record your earnings amount and you will be paid by cash privately.

Fixed Role and Random Group

In this experiment, you will play the role of either a **Seller** or a **Buyer**. Your role will be assigned at the beginning and will remain fixed throughout the entire experiment.

There are **30 rounds** in total. In each round, you will be randomly grouped into a market consisting of **3 participants** (including yourself): **2 Sellers and 1 Buyer**. Groups are **randomly shuffled** every round, so you will likely interact with different people in the room from round to round. You will never know exactly who you are matched with.

Products and Fit Values

Sellers offer products to the Buyer. A seller's product can either be a Good Fit or a Bad Fit for the buyer, with equal chances.

- **Good Fit:** Value to the buyer is **100 points**. This happens with a chance of **50%**.
- **Bad Fit:** Value to the buyer is **0 points**. This happens with a chance of **50%**.

Before visiting a seller, the buyer does NOT know whether the product will be a good fit or bad fit. The fit is determined **randomly** and **independently** for each seller. The buyer only knows a product's fit value ***after visiting*** that seller. Knowing the fit with one seller does not provide any information about the fit with the other seller.

Decisions in Three Stages

Seller Bidding $\Longrightarrow$ Seller Pricing $\Longrightarrow$ Buyer Visit & Purchase

Each round proceeds according to the following three stages:

Stage I: Sellers Bid and Pay for Top Position

The two sellers simultaneously submit bids (in points) to compete for the "Top Position" on the Buyer's screen. A seller cannot see the other seller's bid when choosing their own bid. Bids must be between 0 to 100 points.

- **The Top Seller:** The seller with the higher bid wins and is displayed **first** to the buyer. It costs the buyer **0 points** to visit the **Top Seller**.
- **The Second Seller:** The seller with the lower bid loses and is displayed **second** to the buyer. It costs the buyer **20 points** to visit the **Second Seller**. The buyer decides whether or not to pay this cost, meaning the Second Seller might not be visited at all.

[3]These instructions are for $s = 20$. In the high-search-cost treatment, the search cost is 40 rather than 20 points.

Tie-breaking: If both sellers submit the exact same bid, the computer will randomly select one seller to win and become the Top Seller (each seller has an equal chance).

How the Bidding Payment Works

The **Top Seller** wins the position, but pays a position payment equal to the **Second Seller's bid** (not their own bid). The **Second Seller** pays **0 points**.
Example: Seller A bids **22**. Seller B bids **14**. Seller A becomes the Top Seller, but Seller A only pays **14 points** (not 22). Seller B becomes the Second Seller and pays **0 points**.

Stage II: Sellers Set Prices Observed by Buyer

After sellers see whether they are the Top Seller or Second Seller, they simultaneously post a price for their product. Once posted, both prices are **immediately observed** by the buyer. When setting prices, sellers do not know if their product is a Good Fit or Bad Fit for the buyer.

Stage III: Buyer Visits and Purchases

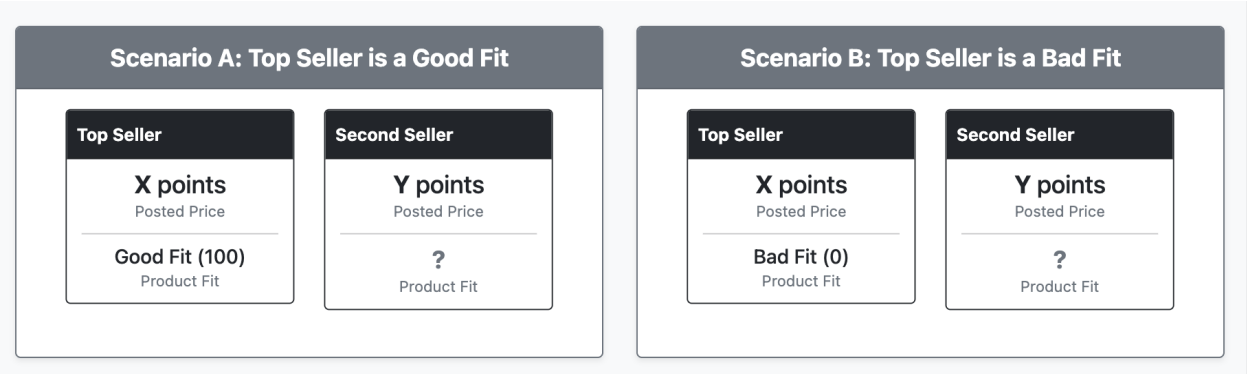
The buyer observes the prices posted by both sellers. The buyer automatically visits the **Top Seller** for free and learns its product fit.

The buyer then decides whether to pay a **Visit Cost** of **20 points** to discover the fit of the **Second Seller**. **The Visit Cost is non-refundable**, even if the buyer ultimately decides not to buy from the Second Seller after the visit.

Finally, the buyer can choose to purchase one product from any seller the buyer has visited. Specifically, a buyer who visits the Second Seller can still choose to buy from the Top Seller, without paying any *additional* visit cost. The buyer may also choose to quit the market at any time and buy nothing.

Examples of Buyer's Decision Screen

For example, if sellers post prices of **X** and **Y**, the buyer observes **one** of these two screens, depending on whether Top Seller is randomly determined to have a Good Fit or Bad Fit. Notice that buyer always knows Top Seller's fit, but Second Seller's fit remains **?** until visited.



Points and Earnings

At the end of the experiment, the computer will randomly select **ONE out of the 30 rounds** to be your "Payment Round." Your points in this Payment Round will be converted into dollar earnings using a conversion ratio of **10 points = $1**.

You will start each round with an initial balance of **200 points** (worth $20), which will be added to your payoff points in that round. This starting balance resets each round and does not carry over. Your total earnings for this experiment will be **$10** show-up fee plus your converted points in your (randomly selected) Payment Round.

Your payoff points are calculated as follows:

Buyer Payoff Points:

- If you purchase: **Product Fit** − **Price Paid** − **Visit Cost** (if you visit Second Seller)
- If you quit without purchase: − **Visit Cost** (if you visit Second Seller)

Notice that Visit Cost is non-refundable for the buyer.

Seller Payoff Points:

- If you make a sale: **Price Charged** − **Position Payment**
- If you do not make a sale: − **Position Payment**

Notice that Position Payment is non-refundable for sellers.

For each round: **Total Points = 200 Points + Payoff Points**

Please be patient while waiting for others to finish their decisions. At the end of each round, you will see a feedback screen showing your payoff and total points for that round.

Below is a summary of the instructions. Next, you will answer three comprehension questions and play two practice rounds. These practice rounds help you learn the interface and do not affect your earnings. After that, we will start the formal rounds. Once you have made a choice, you cannot go back to a previous screen to change it. **If you have any questions, please raise your hand at any time during the experiment.**

Instructions Summary

Rounds	**30 rounds**. Your role is fixed: you are either a **Seller** or a **Buyer**.
Groups	Each round: **2 Sellers + 1 Buyer** are **randomly** grouped.
Three Stages	Bidding $\Longrightarrow$ Pricing $\Longrightarrow$ Visit & Purchase
Bidding	Highest bid becomes **Top Seller** and pays the other Seller's bid.
Pricing	Prices posted by both Sellers are **observed** by Buyer before any visit.
Product Fit	When Buyer visits a Seller, Buyer discovers if the product is a **good fit** worth **100 points** (**50%** chance), or a **bad fit** worth **0 points** (**50%** chance). The fit is determined **independently** for each seller.
Visit Cost	Visiting **Top Seller** is free. Visiting **Second Seller** costs Buyer **20 points**.
$Earnings	Show-up **$10** + Points of **ONE** random round (**10 points = $1**).